

University of Sheffield

Visualising Emotion: A Similarity-Based Analysis of Emotional Fidelity in Text-to-Image Generation

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Declaration

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Abstract

Visitors to Sheffield's Festival of the Mind typed how they felt and a text-to-image installation answered each entry with a monochrome line drawing. That left 1,025 unlabelled prompt–drawing records, of which 1,012 are analysed here. Nobody rated the drawings, so there is no ground truth. The question asked instead is whether one model's reading of a drawing agrees with another model's reading of its prompt. The prompt reader is RoBERTa fine-tuned on GoEmotions. The image reader is describe-then-extract, where BLIP-VQA is asked five factual questions and BART-MNLI reads an emotion family from the resulting description. On the supervisor's advice the readings are scored on a four-rung similarity ladder, from valence–arousal similarity to family agreement, and never matched label to label. Because valence–arousal similarity cannot fall below 0.412, with random pairs averaging 76.6%, every score is reported beside its random-pairing null, and because 249 records repeat a prompt, the intervals and tests are clustered on prompt groups.

On the 607 prompts the prompt reader labels emotional, observed valence-sign agreement is 64.0% over the 530 of them that carry a sign, where random pairing averages 45.6% ($z = +11.1$), and family agreement is 53.0% against 38.8%. Both hold with each prompt counted once and after Holm correction. Under the describer's split, face-classified drawings reach 73.9% on valence sign and scene-classified drawings reach 42.9%, a gap of +30.9 pp (clustered 95% interval [+22.6, +40.6], clustered permutation $p = 0.0001$). Neutral-labelled prompts stay at chance. Direct CLIP was also run like for like; it gets 22.4% family agreement where describe-then-extract gets 53.0%. Where a prompt repeats, 68.8% of its drawings receive the group's modal family, against 59.4% with families shuffled.

The describer reports no person in 414 scene-classified drawings yet gives an expression for 406 of them, and corpus-wide, wherever that answer is an emotion word, the image reader returns that word's family 99.8% of the time. Dropping the expression sentence cuts valence-sign agreement to 15.3% (null 11.4%), and a people-aware template leaves the face rows unchanged and the scene rows at chance, so the headline agreement is a property of the measurement chain. To say what a person sees would need a human-rated study. On the 120 drawings coded blind for this check, the describer's people count is right 87.5% [80.4, 92.3] of the time but the face rule holds a precision of only 42.2% [30.9, 54.4], and half the drawings it calls smiling have nobody in them. The dissertation contributes the corpus and the research question, the null-referenced ladder, the consistency experiment, and an audited, reproducible toolkit and demonstrator.

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Chapter 1

Introduction

1.1 Context

In September 2024 the University of Sheffield held its Festival of the Mind, and at one installation visitors were asked to type in how they felt. Whatever they typed went straight to DALL·E, the model named in each file's provenance data, and the line drawing that came back, in black and white, went up on the screen for the visitor. There are 1,025 of these prompt–drawing records in all. They arrived at the start of the project in June 2026 as a folder of PNG files and a single JSON file of bare strings, without labels or metadata and with nothing to say who had typed what. The project specification offered a text strand and an image strand and left the focus open.

Two features of the material shaped the work. First, the entries are short, and most of them are about the writer. The median length is four words, and 294 of the analysed records consist of one word only. The word "happy" by itself was typed 42 times, and other common entries are "curious", "ok", "excited" and "good". Many entries name more than one feeling at once, often feelings that pull against each other, as in "tired but happy" or "tired proud happy", and "tired" turns up inside many of the longer entries. A single label would lose part of what such an entry says. Second, every drawing is monochrome. The installation software appended a line-art instruction to each prompt (the instruction itself is not part of the data), so colour, which the literature treats as the main carrier of emotion in generated images [2], is missing from all 1,025 drawings. Whatever an automatic reader can recover from them has to come from line, posture, faces and composition.

The interface accepted any text at all. Some entries are not about a feeling at all: "pigs", which appears repeatedly, "mickey mouse", "disney characters", "i like football", and questions put to the machine such as "how can a circle be squared". Nothing ties a record to a person, so someone could have typed once or come back several times, and how many people contributed is unknown. Since 249 of the 1,012 records repeat an earlier prompt, the count of contributors is at most 763, the number of distinct prompts. Every count in this dissertation is therefore a count of records. Figure 1.1 gives six records in the form used throughout this dissertation, the visitor's words with any personal name replaced by a placeholder, next to the drawing made from them.



Figure 1.1: Six records as they appear throughout this dissertation. Under each drawing are the record number, the prompt as typed, the class the describer assigned (face or scene, Section 3.4) and the family returned by each of the two readers. The demonstrator examples (Section 3.9) are records 63 and 201. In record 21 the drawing is of mountains, the describer reported no people, and its expression answer was still "smiling"; Section 4.8 reports how often that happens.

1.2 The question

If someone types something about how they feel and an image generator draws it, how far does the emotion read from the words agree with the emotion read from the drawing? That is the research question, and the data limit how it can be answered. Nobody rated these drawings, and the only record of what a visitor meant is what the visitor typed. Each side therefore gets an automatic reader in place of a person. A language model assigns the prompt an emotion, and a second reader does the same for the drawing by first putting it into words and then reading those words. All the numbers reported here are agreement between those two readings. What a person looking at a drawing would see is a separate question that needs human raters, and Chapter 6 lists that study as further work.

The design follows a direction Dr Chen gave at a meeting at the end of July. Dr Chen is a psychologist who works on facial expressions, and her point was that scoring a drawing as a hit or a miss against a label is too crude for emotion. Two people looking at the same face will often not agree about what it is doing, and the models are noisy on top of that. The labels overlap as well. Cowen and Keltner [19] found 27 categories in self-reported experience, joined to one another by continuous gradients, and on Russell's circumplex [74, 67] "happy" lies a short distance from "excited" on the valence–arousal plane. If a visitor typed "happy" and the drawing is read as excited, counting that as a complete miss throws away most of what this corpus has to offer. So the project asks how similar two readings are, and it never scores a drawing on whether it matches a label. Dr Chen asked for this similarity framing to be named as the novel element of the work.

In place of the hit-or-miss score there are four measures arranged as a ladder, from loose to strict, and each is compared with what comes out when drawings are paired with prompts at random. Chapter 3 defines them. Valence–arousal similarity is the loosest and family agreement, which needs the two top families to coincide, is the strictest; top-3 soft overlap and valence-sign agreement sit between them. The chance reference is needed because the loosest measure does not start at zero. Valence–arousal similarity has a floor of 0.412 on a scale whose top is 1, and a random pairing of drawings with prompts already scores about 76.6%, so a value that looks respectable at first sight can sit below chance. This surfaced while the demonstrator was being built, when a value that looked fine on screen was in fact under the random-pairing mean.

1.3 Aims and objectives

The aim is to measure how closely two automatic readers, one reading the visitor's words and the other the drawing, agree about the emotion in a record, with each value set against its chance level, and to describe the instrument well enough that its weak points are on record. There are eight objectives.

1. Build the image side as a describe-then-extract pipeline, in which a vision model answers factual questions about the picture and a separate text model reads an emotion from those answers, so that the vision model is never asked about emotion.
2. Define the four-rung similarity ladder and the random-pairing null, so that no value

is reported without its chance level.

3. Measure agreement on the full corpus and then separately on the two splits Dr Chen asked for: emotional prompts against neutral-labelled ones, and face-classified drawings against scene-classified ones.
4. Put the face/scene difference to a formal test, and check that the headline results are not an artefact of the repeated prompts.
5. Run a like-for-like comparison between describe-then-extract and a direct reader built on Contrastive Language–Image Pre-training (CLIP), with both given the same records, the same prompt-side readings and the same measures.
6. Turn the repeated prompts into a consistency measurement, asking how far drawings made from one and the same prompt are given the same family by the image reader.
7. Audit the instrument. The checks are a control audit of the neutral-labelled prompts, an audit of the describer's answers, a taxonomy of the strongest disagreements by failure mode, a single-coder check on a blind sample of drawings, and an ablation over the description template.
8. Release the toolkit in a reproducible form, with the headline numbers pinned by tests and a demonstrator that runs the real models.

1.4 Contributions

Very little of the software is new in itself. The prompt side uses RoBERTa fine-tuned on GoEmotions, the image side uses a visual question-answering (VQA) model followed by a natural-language-inference (NLI) model, and the inference relies on textbook procedures such as the clustered bootstrap and the permutation test. The new parts are the object of study and the question put to it, the practice of setting every answer against its chance level, the consistency experiment, and the audits.

The material is new because nobody gave the visitors a list of emotions to pick from or a template to fill in, and what they typed is mostly short self-reports from members of the public, many naming more than one feeling, each drawn as a line drawing with no colour in it. The closest prior study is by Lomas and colleagues [49]. Their design differs on several points: ten emotions from a fixed list, each generated on a person or a robot, in colour, and scored by human raters. None of the 73 papers retained for the review in Chapter 2 measures agreement between the emotion read from a short, mixed self-report and the emotion read from a colour-free drawing made of it.

For each record the question is whether the two readings agree end to end, and it is answered with the ladder set against its null. Because 249 records repeat a prompt and cannot be treated as independent, the intervals are clustered on prompt groups, and the one contrast the design was built around, face-classified against scene-classified drawings, gets a formal permutation test. The repeated prompts also allow a consistency experiment. It needs the same words drawn more than once, which none of the studies reviewed in Chapter 2 had, and it measures how often the image reader gives the same family to a prompt drawn again.

The audits put numbers on the instrument's weak points: how far the image reading depends on a single word from the describer, what the describer answers when the drawing has no face in it, how many of the neutral-labelled prompts contain a plain affect word after all, and how many of the strongest disagreements can be traced to the generator as against the readers. Until the second audit in September the scene drawings looked as if they sat a little above chance. The template ablation in that audit removed that result (Chapter 4).

In practical terms the deliverables are the analysis code, the result tables over the 1,012 records, the demonstrator, a two-layer test suite and the audits themselves. The demonstrator is a small Flask application, run locally, that puts any analysed record or an uploaded image through the real models and reports all four measures against the chance levels the analysis uses. One test layer is a shell script that checks the environment, the corpus files and the cached model outputs, then runs the live pipeline over a few fresh images; the other pins each headline number to the JSON file behind it. Each statistic in the text is a LaTeX macro produced from those same result files. Re-running the analysis rewrites the numbers here without anyone retyping them.

1.5 Overview of the report

The literature review in Chapter 2 goes through eight themes, from theories of emotion to how the public receives pictures made by AI, and closes by stating the gap the project sits in. Chapter 3 covers the corpus and its pre-processing, then the two readers, then the ladder with its null and the inference procedures, then each analysis, the demonstrator and the alternatives tried first and dropped. The numbers, with their tables and figures, are in Chapter 4. Interpretation is held over to Chapter 5, which sets out what the results do and do not establish, why the weakest part of the chain is the describer, and how the findings sit next to earlier work. Chapter 6 closes with further work in the order the evaluation points to. Four appendices follow. Appendix A gives the layout of the repository, the order in which the scripts run and the pinned environment. Appendix B collects the fixed strings the pipeline depends on: the five describer questions, the description template, the hypothesis string, the expression-word list, the category-to-family map, the circumplex centroids, the coder's instructions and the redaction procedure. The record of supervision meetings is Appendix C, and the glossary is Appendix D.

Chapter 2

Literature Review

2.1 Scope and method of the review

Five bodies of work bear on this project, and few papers cite across them. The psychology of emotion, emotion detection in text, image generation from text, the scoring of generated pictures, and the study of how people judge feeling in a picture each have something to say about a drawing made from the words "tired but happy". All five were read across to check that none of them had already asked this project's question. The starting point was the eight papers Dr Chen provided when the project began. From each of those the reference lists were followed outward, one literature at a time. Ninety-seven papers were read in full.

Every paper got two scores, one for how close it sat to the path of the dissertation and one for the strength of its evidence, and a paper was kept only when it did well on both. In practice that meant an empirical study with an evaluation of its own, or a dataset, lexicon or benchmark that later work builds on. Surveys and position pieces went out. So did careful work that was aimed somewhere else, and a reason was logged for every exclusion, so that the 24 papers dropped at this stage can be checked against the criterion. Two foundations would have failed the evidence test applied strictly, Ekman on basic emotions and Russell on the circumplex, and they were kept regardless, because every classifier and every lexicon further down assumes one or the other. That left 73 papers, and they fell into eight themes. On 9 September, after the second audit, 18 references were added for the models and the statistical procedures that the methodology depends on, which takes the bibliography to 91 entries; they are cited where they are used, here and in Chapter 3. The eight themes give this chapter its sections.

One term needs settling before the review starts. When other people's papers discuss how well an image corresponds to its prompt they say *alignment*, and their term is kept here whenever their work is described. For this project the word is *agreement*. What is measured here is how often two readers, one given the prompt and one given the drawing, return the same thing, and calling that alignment would imply a standard that the drawing either meets or misses, and this corpus has no such standard.

2.2 Foundations: emotion theory and taxonomy

The vocabulary comes from two traditions. Ekman [23] made the case that a small number of basic emotions exist, each with its own signal, and the six-way scheme that most classifiers still use is his: happiness, sadness, anger, fear, surprise and disgust. Russell's circumplex [74] puts affect on two continuous axes instead, valence and arousal, and the later paper by Posner, Russell and Peterson [67] argues from neural and developmental evidence that the dimensional picture fits better than the categorical one. Both are used here. The seven families in this dissertation, Ekman's six with neutral added, form the categorical rung, and the dimensional rung comes from where each family sits on the circumplex.

Neither scheme on its own covers what people report. Cowen and Keltner [19] asked people what they felt while watching more than two thousand brief video clips, and the reports resolved into 27 distinct categories linked by continuous gradients; the categories explained more of the variance than valence and arousal did, and neither description was enough by itself. The similarity framing rests on this result. If "happy" and "excited" lie next to each other on a gradient, then a rule that counts a drawing read as excited as a complete miss for a prompt that said happy is really measuring where the boundaries of the label set fall.

The same point has been made for faces in Dr Chen's own field. Using reverse correlation, Jack and colleagues showed that Western and East Asian observers do not hold the same mental representations of the six basic expressions [36]. A later study found that the signals in a dynamic expression arrive in a hierarchy over time, so that the early movements reduce six emotions to four, fear being confusable with surprise and disgust with anger at the start [35]. Dr Chen and her colleagues found distinct facial signatures for pain and for pleasure inside each culture [12]. They then moved psychologically valid expressions onto social robots and a conversational virtual agent by measuring what observers reported seeing in a face [14, 13]. Their most recent work finds that facial movements code the category of an emotion and its intensity as separate components [15].

For a project that reads emotion off drawn faces, this has consequences. There is no reference drawing of an emotion to compare against, and near misses between neighbouring categories are to be expected. What a face signals is settled, in this tradition, by what human observers report, which this project does not measure because both of its readers are programs. So the foundations settled two design decisions, a small family scheme that two quite different readers can both output and a dimensional measure beside it, so that a drawing read as excited for the prompt "happy" earns partial credit.

2.3 Emotion from text

Text emotion recognition has mostly been built and tested on longer social-media text, and it is weakest on short first-person entries like the ones in this corpus. The reference resource is GoEmotions [21]: 58,000 Reddit comments, each annotated with one or more of 27 fine-grained categories or with neutral, together with a mapping from those categories onto Ekman's families that the authors themselves propose and that this project uses. Their BERT (Bidirectional Encoder Representations from Transformers) model's macro-F1

was 0.46. The average hides a lot, because the score falls towards zero on the rare categories whose emotion is implied and never named. Harutyunyan and Kumar [28] re-ran the comparison on the same data across classical learners, a recurrent network and transformers, and the same shape appears in every row. Yang and colleagues [84] used large language models to audit emotion datasets and found short texts under-specified as a rule; the label frequently turns on context that the text leaves out. A festival entry is that case, since it is written in the first person and is a few words long, and it often names two feelings.

Two papers take short texts on directly: Liu and colleagues [48] propose an improved multi-label method for them, and Bhat [5] reports that a small transformer classifies short English texts well. Both were built and tested on tweets, though, and a tweet still gives the model more to work with than "tired but happy" does.

Dimensional resources go part of the way round the problem. EmoBank [8] rated ten thousand sentences on valence, arousal and dominance twice over, once from the writer's point of view and once from the reader's, and found that the two sets of ratings differ, which matters for this project because a visitor's words come from the writer's side and every reader in the pipeline sits, by construction, on the reader's side. The NRC lexicons supply valence, arousal and dominance norms at word level, first for twenty thousand English words [56] and now for fifty-five thousand [57], and the earlier crowdsourced lexicon [60] maps words to basic emotions. Linguistic Inquiry and Word Count (LIWC) [77] is still the transparent tool that psychologists reach for, a validated dictionary whose categories include positive and negative emotion, anxiety, anger and sadness. Zanwar and colleagues [89] found that fusing such psycholinguistic features with a transformer helps it generalise out of domain, which is a live worry for a corpus unlike anything the models were trained on. Niu and Mower Provost [62] compress what a large language model knows about emotion into a compact recogniser that needs no labelled examples. Parkington and colleagues [65] compared a large language model with expert human coders doing thematic analysis, and found high agreement between them but a loss of fine detail on emotional triggers. BERTopic [27] was considered for pulling themes out of the corpus and set aside, because topic modelling has nothing to work with when the median document is four words long.

The rest of the text-side references are the models used in the pipeline. Entailment-based zero-shot classification [88] turns each candidate label into a hypothesis sentence and puts it to a natural-language-inference model, which says whether the input entails it, and BART [42] trained on MultiNLI [79] is the usual choice for that model and the one that reads the descriptions here. T5 [69] recast every task as text-to-text, its instruction-tuned successor [17] extended that, and some of the vision–language models in Section 2.5 are built on the two of them. Everything runs through the Transformers library [81].

On the text side the prompt reader is the least reliable part of the chain, and that shaped two choices. Its label on a four-word entry is uncertain, and every number downstream inherits the uncertainty. Chapter 3 explains why GoEmotions was chosen over an entailment model and a hand-built lexicon for the prompt side; the short version is that it was trained on first-person Reddit comments, the nearest register to a festival entry available. The decision to keep the prompts that the reader labels neutral as a group of their own and analyse them, the specificity check of Chapter 4, comes from this

literature too. A label of neutral on a two-word entry is exactly the kind of judgement Yang and colleagues warn about.

2.4 Datasets, benchmarks and lexicons

Every image-side resource in this literature is photographic or painted, and every one is in colour. EmoSet [86] is the largest, at 118,000 images, each labelled with one of eight emotions and a set of attributes. Machajdik and Hanbury [50] built their affective classifier from colour, texture and composition features drawn from art theory, and they reported how far people agree with one another on the same images as a reference point. The Nencki Affective Picture System [51] is a set of photographs that come with normative ratings of valence and arousal. EMOTIC [39] annotates people in their surroundings, with 26 categories as well as continuous dimensions, and its finding is that what surrounds a person changes which emotion annotators assign to that person. None of these resources contains line drawings, and that domain shift is one of the two reasons for the shape of the image side (the other, Dr Chen's direction, comes up in Section 2.5). A classifier built on any of these resources could be run over the drawings, but its output could not be trusted, so the pipeline describes a drawing first and extracts the emotion from the description afterwards. The direct-CLIP reader is kept as the baseline that the method is compared against.

The art datasets come closest to this corpus because each pairs a picture with a short piece of affective text. ArtEmis [1] gathered 439,000 utterances in which annotators say what emotion an artwork makes them feel and why. ArtELingo [55] took the same paintings into four languages and found that the emotion labels move with the language and culture of the annotator. WikiArt Emotions [59] covers four thousand paintings and gave two results relied on later. Giving an image a title shifts the emotion people report for it. Images with a face in them get much more consistent emotion judgements from annotators than images without one. On the text side, CARER [76] built a tweet corpus by distant supervision with eight emotions, and it is where the widely used "emotion" dataset comes from; the SemEval affect tasks from 2018 [58] and from 2025 [61] fix the shared benchmarks and the protocol for scoring intensity. Then there are the human-preference datasets built for text-to-image models, ImageReward [83], Pick-a-Pic [37] and Human Preference Score [82], each of which records which of a pair of generated images people preferred, without asking about emotion, and trains a scorer on the answer.

2.5 Text-to-image generative models

The drawings were made by a DALL·E model. The original DALL·E [72] was a discrete autoregressive model, the second version [71] decoded images by diffusion, conditioned on CLIP image embeddings, and DALL·E 3 [4] trained on synthetic captions to follow prompts more closely and rewrites what the user typed before generating anything. Which of these made the festival drawings is not recorded. The provenance manifest inside each PNG names the software agent as "DALL·E" and nothing more, with no version, no seed and no copy of the rewritten prompt. So no drawing can be produced again, and nothing about the generator by itself can be separated from the rest of the chain, a

limit that returns in the consistency experiment. The diffusion models that DALL·E sits beside, latent diffusion [73], SDXL [66] and Imagen [75], share a set of well-documented weaknesses: fine detail, rendered text and faces.

The readers used here are from the same few years. CLIP [68] learned to embed images and captions in one space from 400 million pairs of photographs and text, and it has become the default zero-shot image classifier. It was trained on colour photographs and its strength is objects, and for that reason Dr Chen's direction in late July was not to ask it for an emotion. Bootstrapping Language-Image Pre-training (BLIP) [45] learned both captioning and visual question answering by bootstrapping from noisy web data; its VQA head takes an image and a free-form question and returns an answer, and the describer in this project does that five times per drawing. BLIP-2 [44] joins a frozen image encoder to a large language model through a small trainable bridge. OpenCLIP [34] is the open re-implementation of CLIP, and it is the model behind the direct-CLIP baseline. All of these readers are borrowed instruments, picked for what each is good at, and the choice to put a question-answering describer in front of the emotion reader, instead of a classifier, follows from the domain-shift argument of Section 2.4.

2.6 Image evaluation: metrics and frameworks

Automatic scoring of text-to-image models has settled on one question, whether the objects, attributes and relations that the prompt named are present in the picture. CLIPScore [29] is the cosine similarity of the CLIP embeddings for the image and for the text. TIFA [33] turns the prompt into questions and lets a VQA model answer them against the image. Cho and colleagues' Davidsonian Scene Graph [16] orders those questions so that a dependent question is only asked once its premise has been confirmed. GenEval [25] runs an object detector and checks counts, colours and positions. VQAScore [47, 43] puts one yes/no question to a generative vision–language model, takes the probability of "yes" as the score, and on the human ratings of GenAI-Bench it tracks people far more closely than CLIPScore does. The Conditional Fréchet Distance [38] compares a model's output distribution with a reference distribution, conditioned on the prompt. Behind these sit the caption metrics that came first, SPICE [3] and CIDEr [78], and the Fréchet Inception Distance (FID) [30], which scores image quality without looking at the prompt.

How much any of this can be trusted is answered by two studies. Lee and colleagues [41] evaluated text-to-image models on many axes at once and found that the automatic scores agree only weakly with human ratings on several of them, including photorealism and aesthetics. Zhang and Wan [90] put CLIPScore, VQAScore and DSGScore under stress on a set of about a thousand prompts, which is roughly the size of this corpus, and the first and the third reversed the ranking of models when the seed changed and moved materially after a single pixel was perturbed. VQAScore was steadier, though not immune.

A second group of papers looks at the judge instead of the score. Chang and colleagues [11] describe the way experienced prompt artists push a model towards a style with keywords such as "black and white" and "line art", and note that an output drifts towards the model's average look unless it is pushed. Oppenlaender and colleagues [64] report that driving generation from a body pose changes the feeling of the resulting image without anyone intending it, and that uncanny artefacts change how the image is read emotionally.

Carrasco and colleagues [10] asked how closely human raters' labels follow the emotion a generator had been told to produce, scored it with Cohen's kappa and Krippendorff's alpha, and found the agreement much stronger once the labels were reduced to positive and negative. Across ten experiments, Horton, White and Iyengar [32] together with Millet and colleagues [54] find that people rate the same piece of work lower when they are told a machine made it, which constrains the design of any later human study on these drawings. Two benchmarks test the machine judges themselves. Bhattacharyya and Wang [6] found that vision–language models do a poor job of recognising, zero-shot, the emotion an image evokes in a viewer, lean systematically towards positive labels, and change their answer with the order in which the candidate labels are listed. Gao and colleagues [24] report that multimodal models manage valence, fail on arousal and dominance, and run together what the person in the picture expresses and what the person looking at it feels.

No alignment metric measures affect, so none of them could be the instrument here, although VQAScore, which works by asking a question and reading the answer, is the nearest relative of the describer used here. They are also unstable. Because the scores move with the seed and with a single pixel, no value in this dissertation is reported bare, and each is compared with its random-pairing null. The positive lean and the expressed-versus-evoked confusion that the two benchmarks found in machine judges are the failure modes the audits in Chapter 4 check for, and both appear in this corpus.

2.7 The intersection: emotion in generated images

The papers that put emotion into generated images condition on a target and work in colour. EmoGen [85] generates an image from an emotion category by way of an emotion space mapped onto CLIP's, and scores the result with an EmoSet-trained classifier. Make Me Happier [46] takes an existing image and edits it towards a chosen emotion; its authors found positive emotions harder to separate from each other than negative ones, and fine facial detail was a weak point. EmotiCrafter [20] conditions generation on a continuous valence–arousal coordinate. Mehta and Buntain [53] asked whether generators drift towards fear whatever the prompt says, with an image-emotion classifier for the pictures and a text-emotion classifier for the prompts, and that cross-modal design is the nearest methodological relative of the pipeline here. Zhou and colleagues [91] turn an image into affective captions first and classify the evoked emotion from those, so the describe-then-extract idea has a precedent on the recognition side too.

Alvarado [2] had a diffusion model render emotion words and then extracted colour features from what came back, and found that lightness and chroma track emotional meaning in the output more closely than hue does. Only colour features are modelled. Shape and line are not tested, so the study says nothing either way about what a drawing with no colour in it can do. The literature therefore offered no prediction for a corpus without colour, which is one reason it seemed worth studying.

The nearest prior study is by Lomas and colleagues [49]. Twenty-four raters scored 240 images, made by three generators from a fixed list of ten emotions, with each emotion rendered on a person or on a robot, rating alignment from 0 to 10. The generating model was the largest effect, with DALL·E 3 well ahead of Stable Diffusion and of DALL·E 2, and the scores differed sharply from one emotion to the next; emotions rendered on

a person scored higher than the same emotions on a robot. Their prompts were built from a template and their images were in colour. Nobody, in the papers reviewed, has measured affect agreement for self-reports that came straight from the public, mixed feelings included, and were drawn without colour. None of these studies drew the same words twice either, so none could ask how stable a reading is, and the repeated prompts in this corpus make that question possible.

2.8 Emotion perception and human judgement

Machine recognition of emotion in pictures was built on photographs, and it mixes up two different constructs. Yang and colleagues [87] learned visual sentiment with a pair of coupled networks and weak supervision. EmoSet and the other datasets of Section 2.4 supply the labels such systems learn from, and the benchmarks in Section 2.6 show how brittle the resulting judges are. Underneath this is a distinction, formalised by Gao and colleagues [24] and made concrete in Dr Chen's experiments, between what the person in a picture expresses and what a viewer feels on looking at it. A classifier fitted to crowd labels returns a mixture of the two. The pipeline here reads expressed cues only. It works from a drawn expression, a posture and an action. A study with human raters would be reading evoked affect, and the two cannot be pooled. The reverse-correlation methods that Jack and Dr Chen have used [36, 12, 13, 15] show how such a study should be run: start by measuring what observers report seeing, without presupposing it.

The distinction settles the vocabulary used from here on. What the image reader returns is called a reading throughout. Nothing in the computational results says what a person sees, because nobody looked at a drawing until the single-coder check of Chapter 4, and that check is the only point in the project at which somebody does.

2.9 Context

The drawings were made at a public festival. One study of how the public receives machine-made art bears on them directly. In a study of 565 participants, Ragot and colleagues [70] found that an identical artwork is rated lower as soon as it is labelled as made by a machine. Every drawing here was shown to its author as machine-made, so anyone who rates them in future will have to hide that label or counterbalance it.

2.10 The gap

Taken together, these literatures describe an object that none of them has studied. A short self-report holds blended, graded affect, and short first-person text is where emotion classifiers do worst. A generator renders such a prompt fluently but can falter on faces and fine detail, and it may not even draw the words the visitor typed. The strongest evaluation metrics count objects and are unstable in their own right, while the work on emotion in images reads affect mainly through colour, on photographs and paintings, using judges that lean positive and confuse the emotion a depicted person expresses with the emotion felt by the viewer.

Across the 73 papers, no study measures how far the emotion a reader recovers from a first-person self-report that is short, unmediated and mixed in affect agrees with the emotion a reader recovers from a colourless line drawing generated from it. No study asks, either, how stable that second reading is when the identical words are drawn a second time. A compositional metric can pass a line drawing on the question "is there a person?" and the reader can still return sadness for a prompt that read "happy". Lomas and colleagues used a fixed typology and coloured images, whereas the prompts here are whatever somebody typed, blends included, and the drawings are colourless. The conditional generators treat emotion as a control setting; here it is latent in a visitor's words. The visual-emotion recognisers were trained on photographs and paintings, so monochrome line art sits outside their training distribution, and not one of them keeps expressed and evoked affect apart for such stimuli.

The rest of the dissertation asks two questions. The first is how closely two automatic readers agree about the emotion in a visitor's words and in the drawing made from them, measured against chance, and how much of that agreement depends on whether the drawing has a readable face. The second is how stable the reading stays when one prompt is drawn several times. What a person looking at these drawings would see needs people to answer it and is left to further work. The dissertation's contribution is the corpus and the question, a similarity ladder with a chance level under every rung, a consistency experiment that the repeated prompts make possible, and an audited toolkit; it proposes no new metric or classifier.

Chapter 3

Methodology

3.1 Overview and design principles

Three decisions shaped this chapter more than any others. Two came from Dr Chen in late July and one from the demonstrator in August, and they are set out before following the data through the chapter in the order it is processed.

The first is that the comparison is one of similarity. Dr Chen's view was that, with emotion subjective, people in disagreement and models noisy, scoring a drawing as a match or a miss against one label would be too crude, and she asked for the similarity framing to be stated as the novel element. Each reader therefore returns a scored list over one set of seven families, and the two lists are compared on four measures arranged as a ladder, from a forgiving circumplex distance up to a strict test of the top family (Section 3.5). All four rungs are reported together.

The second is that no value is quoted without its chance level. This one was not there from the start. While the demonstrator was being wired up in August it became clear that records paired at random already scored around 76.6% on the loosest measure, since the seven centroids crowd into a small part of the plane and the lowest value the measure can take is 0.412. So 0.71 looks respectable and is under chance. From then on every cell of every table got a random-pairing null (Section 3.6); the clustered intervals and the test of the face/scene contrast came later, in the September audits (Section 3.7).

The third is that the vision model is at no point asked about emotion. This too was Dr Chen's instruction, since CLIP learned from colour photographs and is good at objects. The describer answers five plain questions about what is drawn, a fixed template makes one sentence of the answers, and a second text model reads a family out of that sentence, so both sides of the comparison end up as text (Section 3.4). How much independence that buys is measured in Section 3.8, and it is less than expected. Figure 3.1 shows the chain.

3.2 The corpus

The data are 1,025 records from Sheffield's Festival of the Mind in September 2024. Visitors typed how they felt. The box accepted any text, and each entry produced one drawing for the visitor to look at. Every drawing is monochrome line art, 1024 by 1024 pixels, because

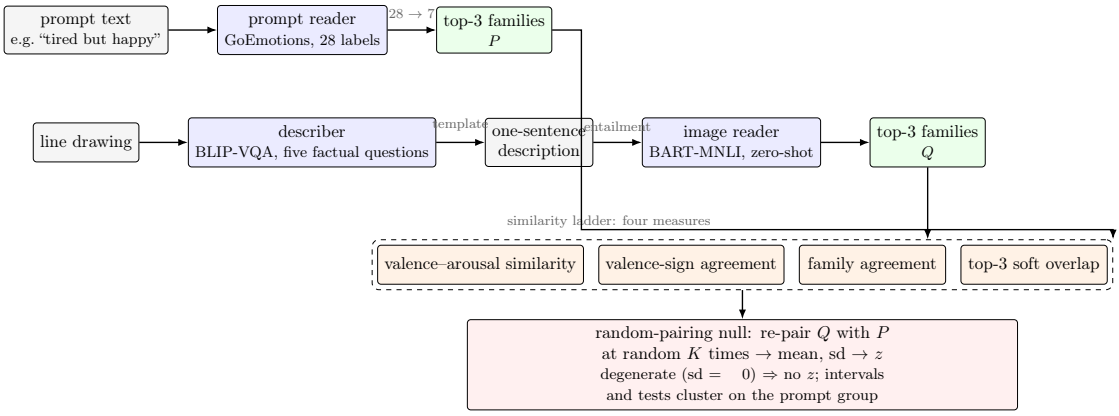


Figure 3.1: The chain from prompt and drawing to the four measures. Top: the prompt reader turns the visitor's words into three scored families. Middle: the describer answers five questions, the template makes one sentence of the answers, and the image reader scores the seven families by entailment. Bottom: the four measures and the random-pairing null that each of them is compared with; a null with zero spread yields no z, and every interval and test clusters on the prompt group.

the installation software added a style instruction to every prompt, and the instruction itself was not passed on with the data. What arrived was one JSON file of 1,025 bare strings keyed by position, no labels, no metadata, no participant identifiers, plus a folder of 1,025 PNG files whose only link to the strings is their order.

Each PNG also carries a C2PA (Coalition for Content Provenance and Authenticity) provenance manifest. It names DALL·E as the software agent, without a version, and holds no seed and no copy of the rewritten prompt that DALL·E composes before it draws [4], so nothing can be regenerated and the generation settings are unknown. Its signing time, though, did a lot of work in the pre-processing (Section 3.3).

Most prompts are short. Half of them have four words or fewer, the mean is about six, the longest runs to 107 words, and 294 of the analysed prompts consist of one word (Figure 3.2). About 78% never use a capital letter, and a handful are written in Italian, in French or in German. The commonest entry is “happy”, drawn 42 times. Mixed feelings are common (“tired proud happy”, “tired but happy”), and a fair number of entries are not about feelings at all: objects (“pigs”), pop culture (“mickey mouse”), questions put to the machine, place names.

Records are not people. Among the 1,012 analysed records, 249 are repeats of a prompt someone had already typed, so how many people took part is unknown, 763 is the upper bound, and any interval that treats records as independent will be too narrow (Section 3.7). Figure 3.3 gives the make-up of the corpus after pre-processing: 13 tests set aside, then the prompt reader's split (607 emotional, 405 neutral-labelled) and, within the emotional prompts, the describer's (402 face-classified, 205 scene-classified).

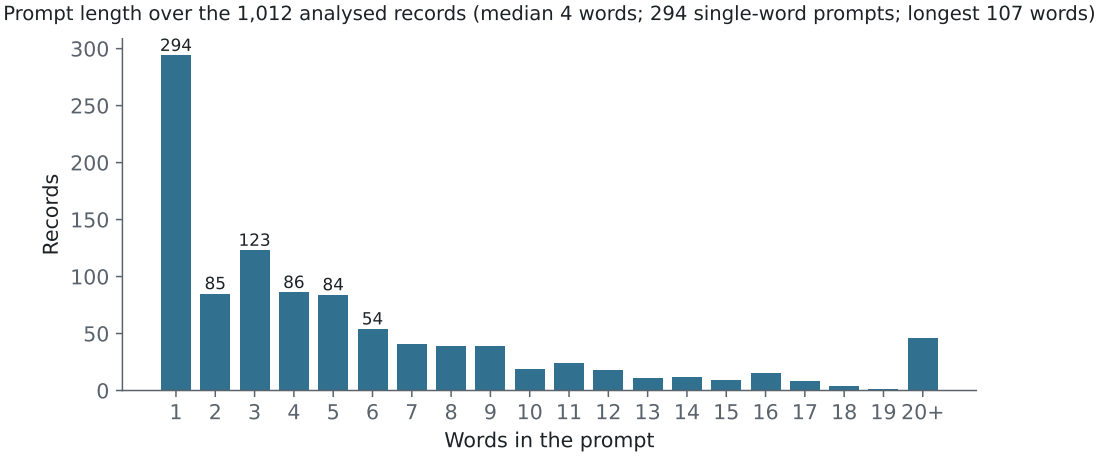


Figure 3.2: Word counts of the 1,012 analysed prompts, with the tail collapsed at twenty words: median four words, 294 prompts of one word, longest 107 words.

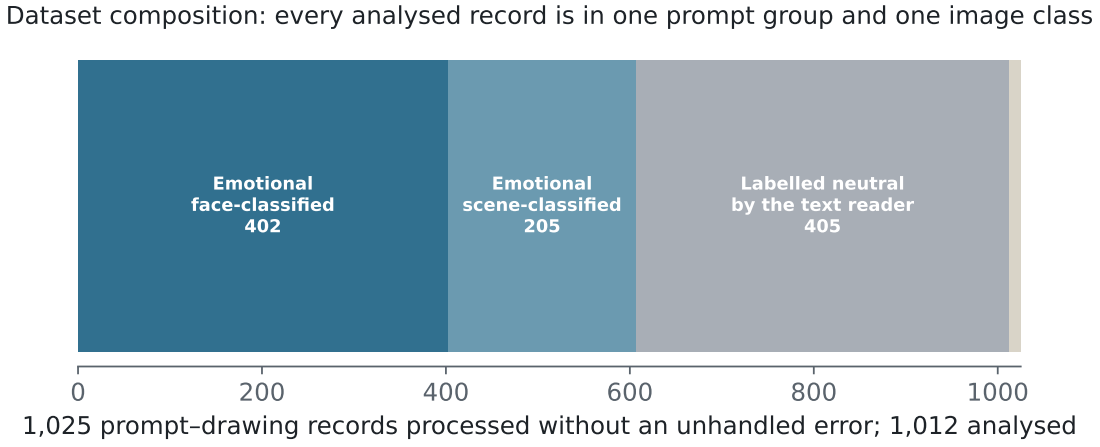


Figure 3.3: What the 1,025 records became after pre-processing: 13 pre-festival tests set aside, then the 1,012 analysed records divided by the prompt reader into emotional and neutral-labelled prompts and, within the emotional prompts, by the describer into face-classified drawings and scene-classified ones. These are groupings for analysis; no analysed record is excluded by them.

3.3 Data pre-processing

3.3.1 Loading and pre-test removal

Nothing could be analysed until the files could be opened. Three naming patterns share one directory. The earliest records are zero-padded (`img_00001.PNG`), the middle block is unpadded (`img_200.PNG`), the records from 1000 onwards have a `t` before the number (`img_t1000.PNG`), and records 2 and 4 end in lower-case `.png`, which a case-sensitive file system reads as another name. A loader written for the first pattern finds fewer than one file in five. `emoviz/data.py` handles all three and asserts that exactly 1,025 records resolve, so a missing file stops the run instead of quietly shrinking the sample.

The timestamps came out of the same loader. Every C2PA manifest carries an RFC-3161 signing time, and reading it gave each image a date the supplied files did not expose (Figure 3.4). The first 13 records date from 26 August to 18 September 2024, before the festival opened, on scattered days, as a developer testing the installation would, so they are excluded from every analysis and pushed through the pipeline all the same, so that the completion count covers everything. All the others up to the penultimate record fall on 19–29 September 2024, with none on 23 September. The last record, signed on 20 October 2024, is kept.

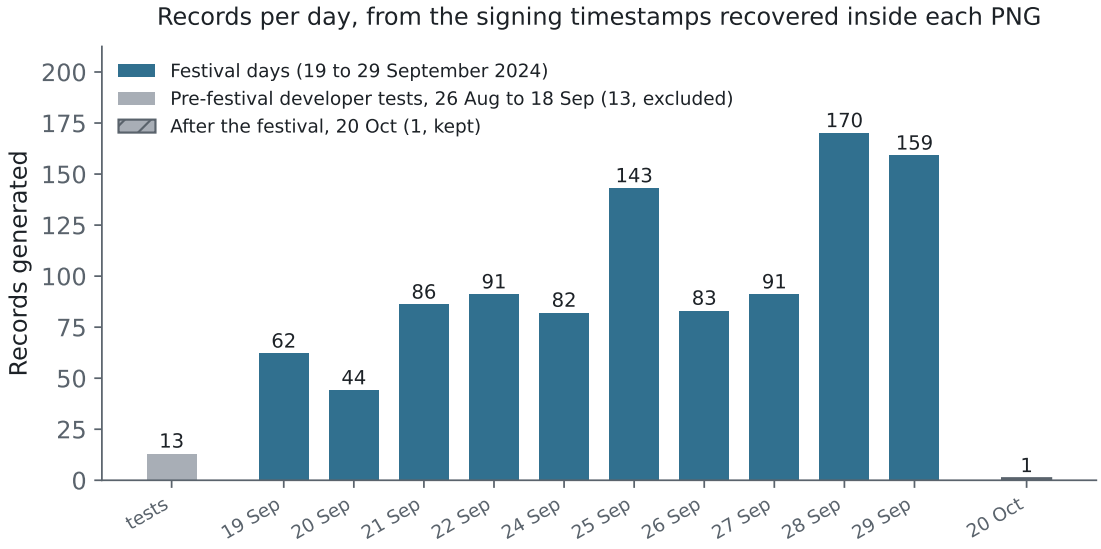


Figure 3.4: Records per day from the signing timestamps inside the PNG files. The 13 developer tests from 26 August to 18 September 2024 are shown apart on the left; the festival days run from 19 to 29 September with none on 23 September; the single record signed on 20 October is on the right and is kept.

3.3.2 Prompt groups

Each record belongs to one prompt group. The group identifier is the prompt with surrounding whitespace removed and case folded. There are 763 groups. 87 contain more than one record, those hold 336 records, and 43 are pairs. The identifier is computed from the text as typed, before any name is redacted, and that order matters, because two different names collapse to one placeholder and an identifier computed after redaction

would merge two unrelated prompts (this went wrong once in development, and the accounting test caught it). The identifier ignores case but the prompt reader did not, so a few groups hold two slightly different readings ("Ok" and "ok" share one); no count changes because of it.

3.3.3 Name redaction

Eight prompts name a private individual, a friend, a relative, a teacher. Dr Chen's direction at the end of July was to handle this as data pre-processing, replacing the name of a private person with a placeholder while leaving in what visitors wrote about illness, grief and hard times, since that content identifies nobody on its own and the analysis needs it. Redaction runs at load time. The files on disk stay as they were, and analysis, figures and demonstrator all read the same redacted text.

The procedure has three stages, each written up in Appendix B. Detection favours recall. Since so few visitors used capitals, three detectors run together, an NLTK list of 7,944 given names [7], capitalised tokens in mid-sentence that are missing from an English word list of 236,000 entries, and phrases of the form "called X" or "my friend X". Review attaches a decision, keep or redact, and a reason to every candidate and ties each redaction to one record, so that "Jack Frost" and "woolly mammoth" stay as they are. The audit refuses to pass while any candidate is undecided or any redaction entry has nothing left to act on. Application then swaps the token and merges adjacent placeholders.

Running the prompt reader again on the redacted text moved every family probability by 0.012 or less and left every top-1 family where it was, so every analysis uses the same prompt labels with or without the placeholder. Two entries were nothing but a name. They reduce to the bare placeholder (2 records), and the control audit (Section 3.8) confirms that they make no difference there either. The decision table names the tokens it removes, so it is kept with the dataset and out of the code folder.

3.4 The pipeline

3.4.1 Prompt reader

The prompt reader is RoBERTa fine-tuned on GoEmotions [21], the `SamLowe/roberta-base-go_emotions` checkpoint, roughly 125 million parameters. Its output is 28 probabilities, one per category. These are added up into the seven families the GoEmotions authors proposed (Appendix B), the seven sums are rescaled so that they add to one, and the top three are kept as the prompt reading. A prompt counts as emotional whenever its top family is anything other than neutral, which makes 607 of the 1,012 analysed prompts emotional and 405 neutral-labelled. Nobody checked those labels. They are the reader's opinion, so the neutral-labelled prompts are kept as a group of their own and audited (Section 3.8).

GoEmotions was the third reader tried on this side. My first pipeline used a hand-built lexicon, transparent but unable to represent "calm but tired" except by letting one word win, and the prompts are full of blends like that. The same NLI model was then put on both sides. On short prompts it over-read surprise. Roughly a third of the corpus was labelled surprise in that run, the hypothesis wording misfiring on two-word inputs.

GoEmotions has no such artefact. It was trained on Reddit comments written in the first person, as close to a festival entry as training data gets, and swapping it in left every headline result where it was.

3.4.2 Describer

The describer is BLIP-VQA [45], the `Salesforce/blip-vqa-base` checkpoint, roughly 385 million parameters. It gets five fixed questions per drawing. They always come in the same order, expression first, then action, posture, people count and scene (Appendix B), and none of the five contains the word "emotion", in keeping with the third principle above. The answers are cached record by record, so the template ablation can later repeat the extraction stage without touching the vision model at all.

BLIP-VQA is what remained after three attempts that went nowhere. The intended describer was Qwen2.5-VL, a vision–language model of 3 billion parameters, and on the 16 GB laptop available it filled the memory, swapped for 4 h 45 min and completed nothing. Florence-2 came next. It failed on the pinned Transformers 5.14 over a `forced_bos_token_id` incompatibility. The plain captioning head of BLIP was tried next, but its captions were so terse that the image reader put almost every drawing into one family, surprise. With BLIP-VQA the full run over the 1,025 images completed, at between roughly 1 and 12 seconds a record, slowing when memory was short.

3.4.3 Description template and image reader

The five answers become one sentence. The template puts the people count and the scene first, "A drawing of {people} showing {scene}.", then the action and the posture, "The person is {action}, {posture}.", and last of all "The facial expression is {expression}." (Appendix B). That sentence goes to BART fine-tuned on MultiNLI [42, 79], the `facebook/bart-large-mnli` checkpoint, roughly 407 million parameters, which scores, for each family, the hypothesis string "This scene conveys {family}" by entailment in the zero-shot manner of Yin and colleagues [88]. The three best-scoring families form the image reading.

One property of the template matters later. The template states, for every record, that there is a person and that the person has a facial expression, even where the describer's own answer to the people question was zero, so a mountain reaches the reader with a person and a facial expression attached. It was written in week one. Nobody measured what that did until the audits in September (Sections 4.8 and 4.10). GoEmotions, the one other reader tried on this side, put almost everything in neutral with nearly all the probability on that family, because it wants feeling words in the first person, which a description of a drawing never has.

3.4.4 Face and scene classes

Dr Chen expected faces to be easier than scenes for any reader and asked for the split. The rule is mechanical. Face-classified means two things are true of the describer's answers, that the people count is at least one and that the expression answer contains, as a whole token, one of the readable expression words in Appendix B, so "man is smiling" qualifies.

Everything else is scene-classified. A digit or a number word in the people answer gives the count. Over all 1,012 analysed records the split is 591 to 421, and inside the emotional prompts it is 402 to 205. The rule was tightened on 1 September, after the first audit, so that only whole tokens count.

The split is partly circular. The class comes out of the describer's answers, and the image reader reads a sentence built from those same answers, which makes part of the face/scene contrast a contrast between two kinds of description. Chapter 4 puts a number on it. The single-coder check, where a sample of drawings is classed by a person who has not seen the describer's answers, is the only place it is broken.

3.5 The similarity ladder

Each reader returns three families with scores, $T = \{(\ell_i, s_i)\}_{i=1}^{3}$ with $\ell_i \in \mathcal{F}$ and $\mathcal{F} = \{\text{joy}, \text{sadness}, \text{anger}, \text{fear}, \text{surprise}, \text{disgust}, \text{neutral}\}$. P is the prompt reader's list, Q the image reader's; $\text{top}(T) = \ell_1$ is the top family. Each family also has $c_\ell = (v_\ell, a_\ell)$, a fixed point on the valence–arousal plane (Appendix B). They were placed by hand from the usual positions of the basic emotions on Russell's circumplex [74]; per-family averages of the NRC valence and arousal norms [56] are the documented alternative, left for further work. The four measures are defined below, beginning with the loosest.

Valence–arousal similarity. A reading is first reduced to one point on the plane, its score-weighted centroid, and two readings are compared by the distance between their points:

$$\mathbf{v}(T) = \frac{\sum_i s_i\, c_{\ell_i}}{\sum_i s_i}, \qquad S_{\text{VA}}(P, Q) = 1 - \frac{1}{\sqrt{2}} \left\| \left(\tfrac{v_P+1}{2} - \tfrac{v_Q+1}{2},\ a_P - a_Q \right) \right\|_2, \tag{3.1}$$

Here (v_P, a_P) is $\mathbf{v}(P)$, likewise for Q, and valence is mapped onto $[0, 1]$ so the two axes span the same range. This rung forgives. A prompt read as joy against a drawing read as surprise scores well here, since the two points lie near each other, although the top families differ. The minimum over the seven fixed points is 0.412 (joy paired with sadness), and random pairings already average about 76.6%, so this measure is never reported without its null beside it.

Top-3 soft overlap. The two readers score on different scales, so each side is rescaled so that its three scores sum to one, $\tilde{p}_\ell = s_\ell / \sum_j s_j$, and the overlap adds up, family by family, the smaller of the two shares,

$$S_{\text{ovl}}(P, Q) = \sum_{\ell \in \mathcal{F}} \min\left(\tilde{p}_\ell,\ \tilde{q}_\ell\right) \in [0, 1], \tag{3.2}$$

It equals one exactly when the two sides coincide. Credit goes to a second or third family that both readers listed, which the top rung ignores, and of the four rungs this is the only one that uses the scores as well as the labels.

Family agreement. For n records the rate is

$$A_{\text{fam}} = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}\big[\text{top}(P_i) = \text{top}(Q_i)\big]. \tag{3.3}$$

Of the four this is the least forgiving, and it is the one other studies most often report. Its chance level depends on the marginals. A reader that answers joy for many drawings will agree by accident with a prompt reader that answers joy for many prompts, and Table 4.11 shows that both readers here do exactly that.

Valence-sign agreement. Let $\sigma(\ell) \in \{+, -, 0\}$ map joy to $+$, surprise and neutral to 0, and the other four families to $-$. Then

$$A_{\text{sign}} = \frac{\sum_i \mathbb{1}\big[\sigma(\text{top}\, P_i) \neq 0 \ \wedge\ \sigma(\text{top}\, P_i) = \sigma(\text{top}\, Q_i)\big]}{\sum_i \mathbb{1}\big[\sigma(\text{top}\, P_i) \neq 0\big]}, \tag{3.4}$$

so a record with an unsigned prompt family drops out of the denominator, and the denominator is printed beside every value it produces; for the emotional prompts it is 530 of 607. It asks the coarsest question there is, whether both readers gave the same sign, and the headline result rests on it.

3.6 The random-pairing null

None of the measures starts at zero. Equation 3.1 is bounded below, and both readers say joy so often (Table 4.11) that top-1 agreement happens by accident. So every value is compared with the same measure after the image readings have been shuffled against the prompts, the exchangeability hypothesis [26]: if a drawing's reading has nothing to do with its prompt, handing each prompt somebody else's drawing changes the measure only by noise. For any subset of n records and any measure m, the image side is permuted K times, π_k being the k-th permutation, and m_k is the measure recomputed on the pairs $(P_i, Q_{\pi_k(i)})$; with $\hat{\mu} = \frac{1}{K}\sum_k m_k$ the mean of these values and $\hat{\sigma}$ their standard deviation,

$$z = \frac{m_{\text{obs}} - \hat{\mu}}{\hat{\sigma}}, \qquad K = 300,\ \text{seed } 7. \tag{3.5}$$

There are two cautions on z. First, 300 re-pairings give a rough estimate of the standard deviation in the denominator. Re-running with seeds 1 to 5 puts the headline valence-sign z anywhere from $+11.2$ to $+12.2$ (the null mean moves by under a tenth of a point). z is therefore used to rank effects and never as an exact probability. Second, a null can be degenerate. When a subset's prompts all have the same top family, shuffling the drawings leaves family agreement exactly where it was, and likewise for the sign when every prompt has the same sign; $\hat{\sigma}$ comes out at zero and z is undefined. The scripts spot this from the spread of the null itself and print nothing in that cell. Before that, two subsets had been flagged by hand and a third missed.

3.7 Inference

Records are not independent, and for a while they were analysed as if they were. 249 records share their prompt with an earlier one, and all the drawings of one prompt inherit whatever that prompt does to each reader. The face/scene permutation test and the interval on that difference were run just before the presentation on 4 September; everything else here, the clustered intervals for every other cell, Cohen's kappa, the Holm correction and the paired tests for the two image readers, was added in the second audit on 9 September, when the marking criteria were revisited. The unit of resampling throughout is the prompt group [9].

Clustered bootstrap intervals. Let $\mathcal{G}$ be the prompt groups in a subset. One draw takes $|\mathcal{G}|$ groups with replacement, pools every record of the groups it drew and recomputes the measure, and the interval is the middle 95% of the B bootstrap values [22], $B = 10,000$, seed 7. A group drawn twice counts twice, so a prompt drawn 42 times moves the interval as one piece of evidence and never as 42.

Clustered permutation test of the face/scene difference. Under the null the image class has no bearing on agreement, so face/scene labels can be shuffled over prompt groups. That is done P times, recomputing the difference each time and counting how often the shuffled difference is at least as extreme as the real one; the two-sided p-value follows North and colleagues [63],

$$p = \frac{k+1}{P+1}, \qquad k = \#\{\text{permuted differences at least as extreme as the observed}\}, \tag{3.6}$$

With $P = 10,000$ the smallest value that can come out is 10^{-4}.

Cohen's kappa. Family agreement is also given as Cohen's kappa, for comparison with other studies,

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \tag{3.7}$$

Here p_o is the top-1 rate actually observed and p_e the rate the two readers' family marginals would produce on their own if the readers had nothing to do with each other [18]. Landis and Koch's bands [40] give the words for it. The interval is the same clustered bootstrap.

Multiple comparisons. 36 z-scores survive the degeneracy filter. Each becomes a two-sided p-value under the normal approximation, and Holm's step-down procedure [31] controls the family-wise rate: with the p-values sorted, $p_{(1)} \le \cdots \le p_{(m)}$, rejection proceeds in order and stops at the first $p_{(i)} > \alpha/(m - i + 1)$, at $\alpha = 0.05$. For the largest z the normal approximation is only nominal, and no decision changes because of it.

Paired comparison of two image readers. Each record gives each reader a yes or a no. The comparison is therefore paired. Let b be the number of records where only the first reader agrees and c the number where only the second does. McNemar's exact test [52] on (b, c) assumes independent records and is therefore optimistic here, so a clustered

alternative sums the differences inside each prompt group and flips the sign of whole groups at random, 20,000 times; the interval on the rate difference is again a clustered bootstrap.

Wilson intervals. The single-coder check produces proportions from a small sample, some of them near 0 or 1, so they get Wilson score intervals [80], which behave where the normal interval breaks down.

3.8 Analyses

The list below follows the order of Chapter 4. Each analysis is one script writing one JSON file, and every table, figure and number in the text is generated from those files (Appendix A).

Splits and the main table. `analysis_final.py` produces the main table, four measures each with its null, for the emotional prompts, for the face-classified and the scene-classified records inside them, for the positive and the negative prompts, and for the neutral-labelled prompts, all by Equation 3.5. Only valence–arousal similarity has a live null in the single-valence cells. The same script repeats the table on distinct prompts only, keeping the first record of each of the 763 groups, and that is the robustness check for repeated prompts.

The face/scene difference. `significance.py` runs the permutation test and the bootstrap interval for the difference, for valence sign and for family. Then `inference.py` supplies the intervals for the remaining cells, kappa for the emotional, face and scene cells, the Holm step-down over the printed z-scores, and the seed check.

Specificity check. Neutral-labelled prompts are the specificity check: their agreement should sit at the null. Only valence–arousal similarity can be read there, since on a subset with one prompt label the other three nulls have no spread, and the analysis reports the degeneracy in place of a z.

Method comparison. `method_comparison.py` changes the image reader and nothing else, with an identical prompt reader, identical records (607 of them), identical measures and identical re-pairings. The second reader is direct CLIP [68, 34]. It is OpenCLIP ViT-B/32 with LAION-2B weights, given the seven family names as text prompts, and its readings sit frozen in a file the main pipeline cannot overwrite. A first-build comparison had used NLI on the prompt side, another subset and another overlap formula, and the first audit on 1 September found its direct-CLIP overlap computed over the wrong label space. Table 4.7 keeps it as a record only; the like-for-like version dates from 9 September.

Consistency across repeated prompts. For a repeat group g of $n_g \geq 2$ records, write $c_g = \max_\ell n_{g,\ell}/n_g$ for the share of its drawings given the group's commonest image family. The consistency C is the mean of c_g over the 87 groups. A pair cannot score under 0.5, and 43 groups are pairs, so the null shuffles image families over all 1,012 analysed records,

keeps the group boundaries where they are and recomputes C, 10,000 times, and the number of groups with $c_g = 1$ gets the same treatment. The scripted null is a 9 September addition. C bounds the reproducibility of prompt, generator and reader together; it cannot isolate the generator, whose settings were never recorded, and one visitor entering "happy" twice looks identical to two visitors entering it once each.

Per-family agreement. `audits.py` also breaks family agreement down by prompt family, as the fraction of each prompt family's records that the image reader put in the same family, with clustered intervals; Table 4.9 has the full cross-tabulation.

Control audit. The same script scans the 405 neutral-labelled prompts for any of 66 ordinary affect words, and it picks out the 2 records that redaction reduced to the bare placeholder. The specificity check is then re-run without either set.

Describer audit and copy rate. The describer's people count is tabulated against whether an expression word is present, and the smile answers are counted with their denominators. The copy rate concerns one-word expression answers. It is the fraction of those records on which the image reader's top family coincides with the family of that word; the strict version admits only words that name a feeling (happy, sad, angry and the like, 418 records) and the broad version adds facial actions such as smile and crying (997 records).

Error taxonomy. `error_taxonomy.py` collects the emotional records on which the two readers disagree about the top family while each gives its own top family a score of at least 0.8. Fewer than the intended 40 qualified, so all were examined. An automatic tagger first marks each record (people count, bare expression word, short prompt, ambiguous raw label, connective), and then every drawing was opened and one primary failure mode chosen out of six fixed ones: (a) an invented face or expression, (b) a misread one, (c) a fair description the reader mis-inferred from, (d) a mixed or ambiguous prompt, (e) a drawing unrelated to its prompt, and (f) a prompt the prompt reader mislabelled. A fixed tie-break rule, printed with the counts, settled records where more than one applied.

Template ablation. `ablate_nli.py` takes the cached answers, builds each description again with a modified template, sends only that sentence back through the image reader, and then computes the ladder and the nulls exactly as the main analysis does, seed included. There are five single-clause deletions (expression, action, posture, people, scene), one variant that sets every expression to "unclear", and two people-aware variants that shorten the sentence for records with a zero people count, one keeping the action and posture clauses and one keeping only the scene clause. A control run with the original template separates numerical noise from real template effects. It changed 0 top-1 readings.

Single-coder check. `human_check.py` draws the sample and scores the coding. The sample is 120 drawings chosen by seed in three strata defined by the describer's output, 40 from the face-classified records, 40 from scene-classified records with a zero people count, and every record whose people answer was vague or a single word, the remainder split

equally between the first two strata. Six numbered contact sheets hold the drawings and nothing else. The author is the coder, filling the four columns in Appendix B blind to prompt, description and readings. Once coded, the script scores the people count and the precision and recall of the face rule, both against the coder, what lies behind the "smiling" answers, and the coder's valence against the image reader's, with Wilson intervals, and then recomputes the face/scene difference on the coder's classification, the one estimate free of the describer's split. The coding was not finished at the time of writing. Figure 3.5 shows the first sheet.



Figure 3.5: The first of the six contact sheets for the single-coder check. Only the drawings and their sheet numbers are shown; the prompt, the description and both readings stay hidden until the coding file is complete.

3.9 The demonstrator

The pipeline also runs as a local Flask application, `app.py` with one page in `web/index.html`, on port 5050. It applies the real models to any analysed record or an uploaded image and shows the five answers, the sentence built from them, both readings, and the four measures against the null the analysis used, computed when the server starts from the result files. Three choices in the interface come straight from the methodology. The axis for valence–arousal similarity begins at 0.412, its floor, and the

middle half of the random pairings is shaded across it, so a below-chance value sits visibly under the band. The verdict puts valence sign and family first and gives the loosest measure only as a percentile among the random pairings. A degenerate chance level is named as degenerate, with no number. The page stores nothing that is typed or uploaded, the request log is switched off, and it was never used to collect data.

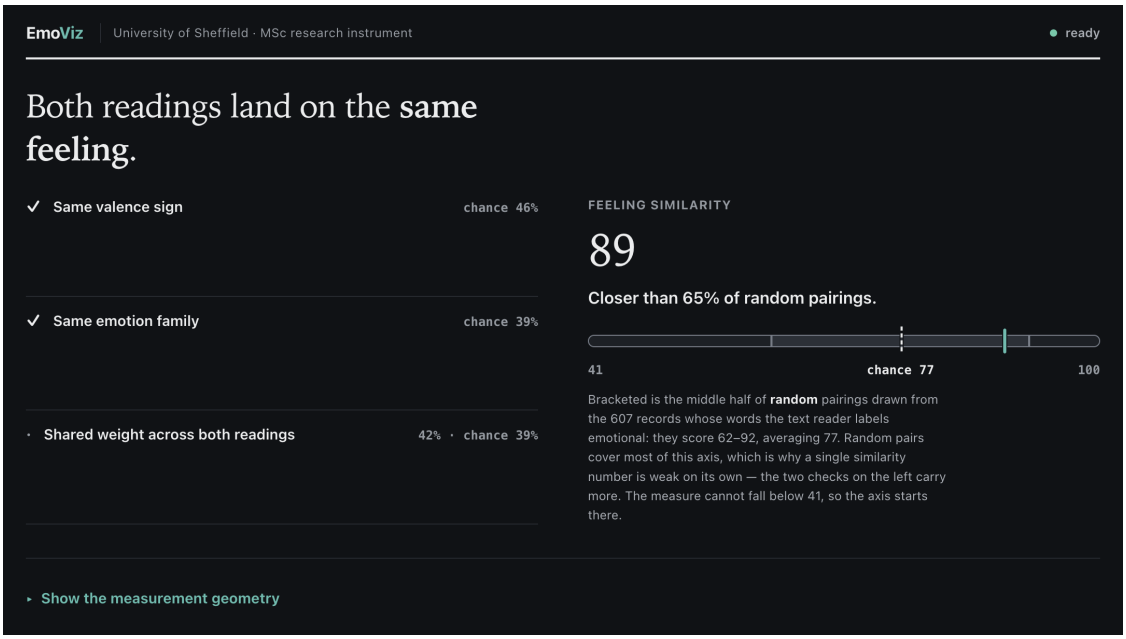


Figure 3.6: The demonstrator's verdict after running record 63, "super tired", through the live models. Valence sign and family are reported first, each beside the chance level for that measure, and the third row gives the top-3 overlap. Valence–arousal similarity is not scored as a check but placed on its own axis on the right: the axis starts at its floor of 41, the shaded bracket is the middle half of the random pairings, the dashed mark is their mean and the solid mark is this record.

3.10 Reproducibility and testing

The environment is pinned in `requirements.txt`, Python 3.12 with PyTorch 2.13 and Transformers 5.14, on Apple silicon in float16, with about 3.5 GB of model downloads on first use. Every random process is seeded. Model outputs are cached record by record, the pipeline checkpoints to a JSONL file after each record so that a crash resumes where it stopped, and every stage after the pipeline reruns in seconds. Direct CLIP's readings are frozen in a file.

A single module, `emoviz/metrics.py`, defines each measure, each constant, the split rule and the null exactly once, and the scripts import from it, so a formula has nowhere to drift. Before it existed, two scripts had their own overlap formulas. They differed slightly. The difference changed no result on this data.

Testing has two layers. The first, `test.sh`, is a shell smoke suite: it verifies the environment, checks that the corpus and the cached outputs are intact, runs the redaction audit, runs each analysis script and requires a clean exit, pushes three fresh images through the live pipeline, and builds the figures, and any failure gives a non-zero exit. The second,

`test_numeric.py`, pins the headline numbers, comparing each one quoted in the text with the JSON value it comes from, to a stated tolerance, and it also unit-tests the measures themselves, the 0.412 floor, symmetry, overlap bounds and scale invariance, the sign map and the face rule. Figures come from `make_figures.py` and the tables and macros from `export_tex.py`, both reading the same JSON, so no result in this document was typed by hand.

The tests also have a limit. None of them compares a model's answer with a person's judgement. The completion figure, every one of the 1,025 images run through without an unhandled error, says that the code did not crash and nothing about whether its answers are any good, and a person enters the chain at one point only, the single-coder check. Appendix A has the file list and the run order.

Chapter 4

Results

This chapter reports the results in the order Section 3.8 introduced the analyses. Interpretation is kept for Chapter 5. Every statistic in this chapter is taken from the analysis result files, and the tables are built from the same files. Each rate is set beside the random-pairing null for its own subset, built by re-pairing image readings and prompts at random 300 times under seed 7, and each interval is a 95% clustered bootstrap over prompt groups with 10,000 draws.

4.1 Overall agreement

Table 4.1 is the main table. It holds the four measures for the 607 emotional prompts, and Figure 4.1 plots each of those cells as a distance from its null, in units of the null's standard deviation. Every rung of the ladder sits above chance. On valence sign the rate is 64.0% where random re-pairing gives 45.6% ($z = +11.1$, 95% confidence interval (CI) [58.6, 68.8]). Its denominator is 530 of the 607, because a prompt whose top family is surprise has no valence sign. Family agreement comes to 53.0% against 38.8% ($z = +10.3$, CI [47.3, 58.7]) and top-3 soft overlap to 50.6% against 39.4% ($z = +11.6$). Valence–arousal similarity is 83.8% against 77.3% ($z = +12.2$). So the loosest measure gives the largest z of the four, and the smallest gap in points between observed and null. Because of the floor in Equation 3.1, random pairs already score high on this measure and the null barely varies, so a few points above it come to many multiples of the null's spread.

4.2 Face-classified versus scene-classified drawings

Figure 4.2 and Table 4.2 divide the same 607 prompts by the class the describer's answers assign to the drawing. Faces are the larger class, with 402 records. There valence-sign agreement reaches 73.9% where the null is 53.5% ($z = +10.4$, CI [68.3, 79.2]), and family agreement reaches 61.9% where the null is 45.8% ($z = +10.3$). The 205 scene-classified records come out lower on both rungs, at 42.9% on valence sign (null 29.6%, $z = +5.8$, CI [33.1, 51.1]) and 35.6% on family (null 26.2%, $z = +4.4$). In this table both classes clear their nulls. The scene row comes up again in Section 4.10.

Each class has a null of its own. For that reason the contrast is stated as a difference in percentage points and tested with the clustered permutation procedure from Section 3.7,

Table 4.1: Agreement between the prompt reader and the image reader on the 1,012 analysed records, by prompt group and image class, against the random-pairing null (300 re-pairings, seed 7). CI: clustered bootstrap over prompt groups (10,000 draws). n/a: degenerate null (every prompt in the subset shares one label, so permutation cannot change the score).

Subset / measure	Observed	Null	z	95% CI
Emotional prompts (n=607; sign n=530)				
Valence–arousal similarity	83.8%	77.3%	+12.2	[82.0, 85.5]
Valence-sign agreement	64.0%	45.6%	+11.1	[58.6, 68.8]
Family agreement	53.0%	38.8%	+10.3	[47.3, 58.7]
Top-3 soft overlap	50.6%	39.4%	+11.6	[46.5, 54.7]
face-classified (n=402; sign n=360)				
Valence–arousal similarity	84.2%	76.4%	+11.2	[82.3, 86.1]
Valence-sign agreement	73.9%	53.5%	+10.4	[68.3, 79.2]
Family agreement	61.9%	45.8%	+10.3	[55.8, 68.0]
Top-3 soft overlap	53.4%	40.4%	+11.6	[49.2, 57.7]
scene-classified (n=205; sign n=170)				
Valence–arousal similarity	82.9%	79.1%	+6.0	[80.4, 85.2]
Valence-sign agreement	42.9%	29.6%	+5.8	[33.1, 51.1]
Family agreement	35.6%	26.2%	+4.4	[26.4, 44.1]
Top-3 soft overlap	44.9%	38.1%	+4.9	[38.7, 51.0]
Neutral-labelled prompts (n=405; sign n=0)				
Valence–arousal similarity	76.6%	76.6%	+0.3	[75.6, 77.6]
Valence-sign agreement	0.0%	0.0%	n/a	degenerate
Family agreement	28.6%	28.6%	n/a	degenerate
Top-3 soft overlap	31.6%	31.6%	+0.0	–

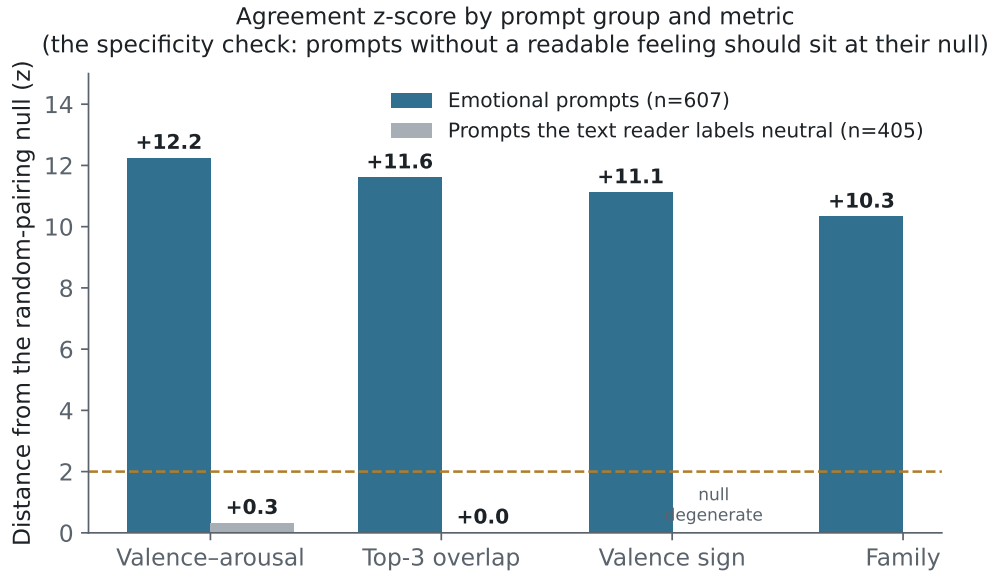


Figure 4.1: Distance from the random-pairing null, in standard deviations of that null (z), by prompt group and by measure. The neutral-labelled group has a degenerate null for valence sign and for family, so those two cells are left empty.

which gives face minus scene as +30.9 pp for valence sign (clustered interval [+22.6, +40.6]) and +26.3 pp for family (interval [+18.4, +35.3]). The permutation test returns $p = 0.0001$ for valence sign and $p = 0.0001$ for family, the smallest value it can report with the number of permutations run. Valence–arousal similarity hardly moves between the classes (84.2% for faces, 82.9% for scenes), so the gap between them sits in the two categorical rungs.

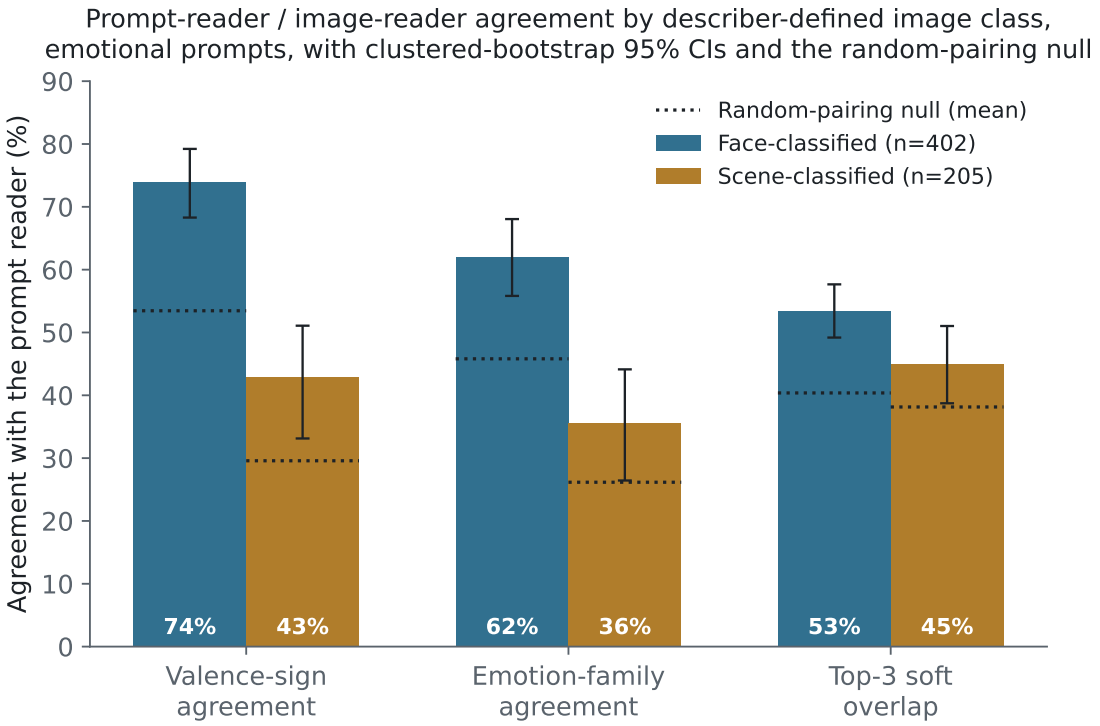


Figure 4.2: Agreement of the image reader with the prompt reader on the emotional prompts, split by the describer-defined class of the drawing. Error bars are clustered-bootstrap 95% intervals. The dotted line in each cell is that cell's random-pairing null.

Table 4.2: Face-classified versus scene-classified drawings on the emotional prompts: agreement with 95% clustered-bootstrap CIs, the difference with its CI, and the clustered permutation p (10,000 draws; significance.py).

Measure	Face (n=402)	Scene (n=205)	Difference (pp)	Perm. p
Valence-sign agreement	73.9% [68.3, 79.2]	42.9% [33.1, 51.1]	+30.9 pp [+22.6, +40.6]	0.0001
Family agreement	61.9% [55.8, 68.0]	35.6% [26.4, 44.1]	+26.3 pp [+18.4, +35.3]	0.0001

4.3 Distinct prompts and the multiple-comparison correction

Because 249 records repeat an earlier prompt, Table 4.3 re-runs the main table with only each prompt's first occurrence kept. That leaves 763 records, 452 of them emotional. The overall and the face results hold: the 319 face-classified records among the distinct prompts give 71.8% on valence sign ($z = +8.4$) and 59.9% on family ($z = +8.1$). The

scene row loses a good part of its margin. For the 133 scene-classified records among them, valence sign drops to 37.3% ($z = +3.5$) and family to 30.8% ($z = +2.6$).

Table 4.4 runs Holm–Bonferroni over all 36 of the z-scores that are not degenerate. 26 pass at the family-wise 0.05 level and 10 fail. The survivors include the full-sample cells of the emotional subset and of both image classes, and the distinct-prompt cells of the emotional subset and of the face class, so every cell reported so far survives except one. Most of the failures were expected to sit at chance anyway: the neutral-labelled control in both blocks, and the valence–arousal and overlap cells of the two single-valence subsets. The one failure among the reported cells is the scene family cell of the distinct-prompt block at $z = +2.6$. It was expected to be informative, and is treated as suggestive only.

Table 4.3: The same analysis on the 763 distinct prompts only (first occurrence of each prompt text kept), a check against pseudo-replication from the 249 repeated prompts.

Subset / measure	Observed	Null	z	95% CI
Emotional prompts (n=452; sign n=394)				
Valence–arousal similarity	83.0%	76.5%	+10.6	[81.5, 84.4]
Valence-sign agreement	62.2%	44.4%	+9.9	[57.2, 67.0]
Family agreement	51.3%	37.2%	+9.3	[46.7, 56.0]
Top-3 soft overlap	48.9%	37.7%	+10.8	[45.9, 51.9]
face-classified (n=319; sign n=284)				
Valence–arousal similarity	83.5%	75.5%	+9.7	[81.7, 85.2]
Valence-sign agreement	71.8%	51.0%	+8.4	[66.4, 77.0]
Family agreement	59.9%	43.0%	+8.1	[54.2, 65.2]
Top-3 soft overlap	51.8%	38.4%	+9.7	[48.1, 55.4]
scene-classified (n=133; sign n=110)				
Valence–arousal similarity	81.8%	78.9%	+3.6	[79.5, 83.8]
Valence-sign agreement	37.3%	27.1%	+3.5	[28.1, 46.2]
Family agreement	30.8%	24.1%	+2.6	[23.3, 38.4]
Top-3 soft overlap	41.7%	36.7%	+3.1	[36.9, 46.5]
Neutral-labelled prompts (n=311; sign n=0)				
Valence–arousal similarity	76.6%	76.4%	+1.1	–
Valence-sign agreement	0.0%	0.0%	n/a	degenerate
Family agreement	27.7%	27.7%	n/a	degenerate
Top-3 soft overlap	31.4%	31.1%	+0.8	–

Table 4.4: Multiple-comparison control over the 36 non-degenerate z-scores the analysis prints (12 degenerate ones excluded). Two-sided p by the normal approximation; Holm–Bonferroni at family-wise 0.05: 26 survive, 10 do not.

Outcome	Tests
Do not survive	DISTINCT negative prompts overlap z+2.7; DISTINCT scene images family z+2.6; FULL positive prompts overlap z+2.3; FULL negative prompts va z+2.1; DISTINCT positive prompts overlap z+1.9; DISTINCT negative prompts va z+1.6; DISTINCT non-emotional va z+1.1; DISTINCT non-emotional overlap z+0.8; FULL non-emotional va z+0.3; FULL non-emotional overlap z+0.0
Survive	the remaining 26, including every headline (emotional, face and scene on all four measures; distinct-prompt emotional and face)

4.4 Specificity check: prompts the prompt reader labels neutral

The specificity check uses the 405 prompts that the prompt reader labelled neutral. Only one measure can move here. Valence sign has a numerator and a denominator of zero on this subset, and the family rate is exactly its null, 28.6%, because every prompt has the same label and re-pairing the drawings cannot change how often the two labels coincide. Valence–arousal similarity does have a live null, and the observed value sits on it: 76.6% against 76.5%, with $z = +0.5$ at record level and -0.1 when the null is clustered on prompt groups.

Table 4.5 reports the control audit. 68 of the 405 (16.8%) contain at least one word from a fixed lexicon of 66 everyday affect words, the commonest being calm (10), content (8), tired (7), stress (6), relaxed (6), inspired (4), sleepy (4), interested (3), anticipation (3), nostalgic (3), and a further 2 consist of nothing but the redaction placeholder. Removing them leaves a cleaned control of 335 records. On those the value is 76.6% against 76.6% ($z = +0.0$), the same as before. The null's standard deviation here is only 0.15%, because the prompt-side points hardly vary between neutral prompts, so observed and expected were always going to land close together. The check passes, and weakly, for that reason.

Table 4.5: The specificity check on the prompts the text reader labels neutral. Only valence–arousal similarity is informative there (sign and family are degenerate: every prompt shares the neutral label). Cleaned control = minus the 2 records reduced to '[name]' and the 68 containing one of 66 plain affect words.

Control set	n	VA observed	VA null	z (record null)	z (clustered null)
All neutral-labelled prompts	405	76.6%	76.5%	+0.5	-0.1
Cleaned control	335	76.6%	76.6%	+0.0	-0.6

4.5 Two image readers, like for like

Figure 4.3 and Table 4.6 score the two image readers under identical conditions: one prompt reader, the same 607 emotional records, the four measures and one set of random re-pairings. On family agreement describe-then-extract reaches 53.0% and direct CLIP 22.4%. The difference of +30.6 pp has a clustered interval of [+25.5, +35.6], the cluster sign-flip test puts p at <0.001, and the exact McNemar test, which takes every record as independent and so errs on the optimistic side, gives $p = 3.4 \times 10^{-35}$. Valence sign goes the same way, 64.0% against 31.1% (+32.8 pp, CI [+27.4, +38.2]), as do overlap (50.6% against 26.4%) and valence–arousal similarity (83.8% against 73.3%). Counting records, 218 of the 607 are agreed on by describe-then-extract alone, 32 by direct CLIP alone, 104 by both readers and 253 by neither.

Two numbers in the table need a note. Direct CLIP's family z is +11.1, above the +10.3 of describe-then-extract, even though its rate is far lower. The reason is that its null sits lower and varies less. Direct CLIP returns neutral for 283 of the 607 images, and an image read as neutral cannot agree by family with any emotional prompt the re-pairing hands it, so chance agreement is rare for it too. A z measures a reader's distance from its

own chance level, so the comparison between the readers rests on the paired tests on the rates above. The second number is the two image readers' own agreement, only 32.1% on top-1 family. Table 4.7 is the first build's version of this comparison, run on 608 records with the entailment model on the prompt side too and an earlier overlap formula, so its figures are not comparable with those of the main analysis. It is kept as history only and none of its figures is quoted.

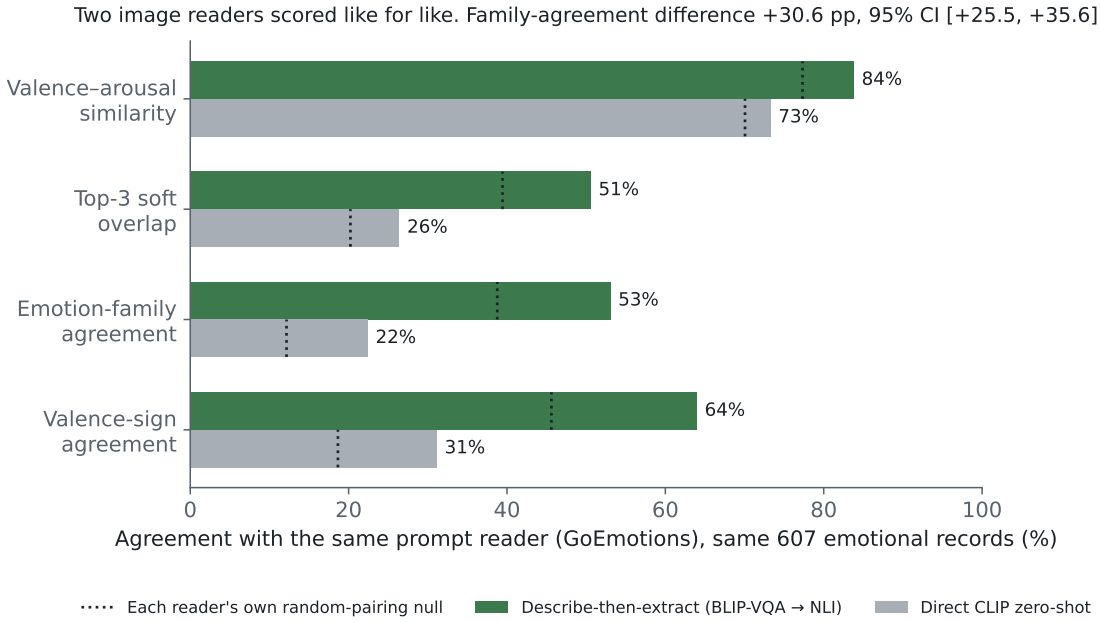


Figure 4.3: The two image readers scored against one prompt reader on the same 607 emotional records. The dotted mark in each cell is that reader's own random-pairing null.

Table 4.6: Two image readers scored like for like against the same prompt reader (GoEmotions) on the same 607 emotional records with the same measures and the same 300 re-pairings. b = records only describe-then-extract agrees on; c = records only direct CLIP agrees on. The exact McNemar p treats records as independent and is optimistic (3.4×10^{-35} for family); the cluster sign-flip randomisation (20,000 draws) respects prompt groups.

Measure	Describe→extract	null	z	Direct CLIP	null	z
Valence–arousal similarity	83.8%	77.3%	+12.2	73.3%	70.1%	+10.2
Valence-sign agreement	64.0%	45.6%	+11.1	31.1%	18.6%	+9.9
Family agreement	53.0%	38.8%	+10.3	22.4%	12.2%	+11.1
Top-3 soft overlap	50.6%	39.4%	+11.6	26.4%	20.2%	+11.2
Paired family test	b=218, c=32			diff +30.6 pp	CI [+25.5, +35.6]	cluster $p < 0.001$
Paired valence-sign test	b=204, c=30			diff +32.8 pp	CI [+27.4, +38.2]	cluster $p < 0.001$

4.6 Consistency across repeated prompts

Repeated prompts form 87 groups, which cover 336 records. 43 of them are pairs. Figure 4.4 and Table 4.8 report how often the drawings in a group get the same reading. Averaged over groups, the share of a group's drawings given its modal family is 68.8%, with a

Table 4.7: The earlier (v1) comparison as published in the build log: prompt side read by BART-MNLI, the 608 records of the v1 lexicon rule, and a different overlap formula. Kept as a record; not comparable with Table 4.1.

Measure	Describe→extract	Direct CLIP
Family agreement	49.3%	25.3%
Top-3 soft overlap	48.6%	22.8%
Valence–arousal similarity	83.7%	74.2%

bootstrap interval over groups of [63.8, 73.9]. The permutation null shuffles image families over every analysed record and re-cuts the same groups, and it gives 59.4% with a standard deviation of 2.1%, so $z = +4.5$ and the permutation p is 1.0×10^{-4}. On valence sign the within-group share is 74.5% against 63.7% ($z = +5.0$). Groups in which every drawing received one family, the fully consistent ones, number 28 of 87 (32.2%), where 14.5 would be expected by chance ($z = +4.2$). For the 43 pairs the share can only be one half or one, which puts a floor under the mean.

The largest group is the single word "happy", drawn 42 times. Of those drawings the image reader returned joy for 34, neutral for 7 and surprise for 1, and Figure 4.5 lays out a selection of them with the reader's family printed under each.

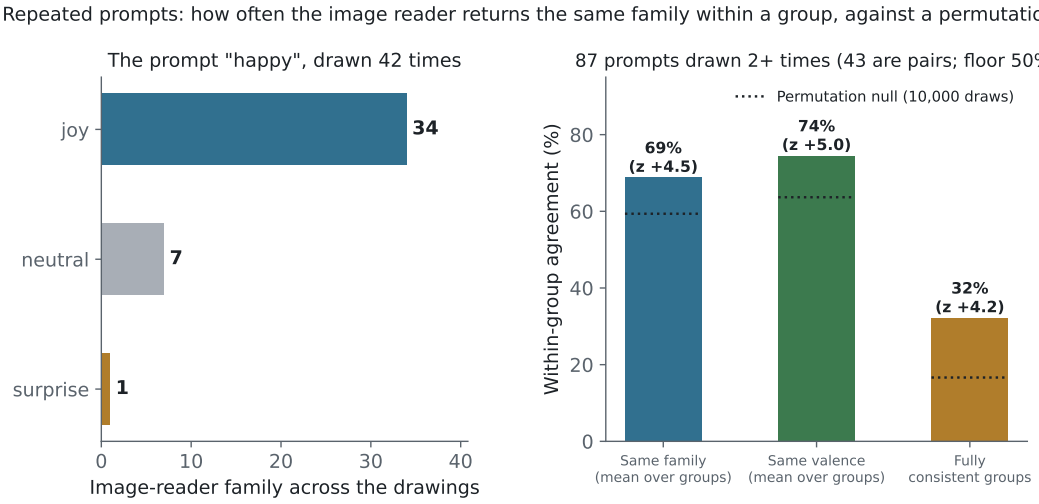


Figure 4.4: Repeated prompts. Left: the image reader's top-1 family over the 42 drawings of "happy". Right: for all 87 repeated prompts, the share of a group's drawings given its modal family, with the permutation null.

Table 4.8: Within-group agreement of the image reader across the 87 prompts drawn two or more times (336 records; 43 groups are pairs, whose modal agreement cannot fall below 50%), against a permutation null that shuffles the image readings over all 1,012 analysed records and re-cuts them into the same groups (10,000 draws, seed 7).

Quantity	Observed	Null mean (sd)	z	p
Same family, mean over groups	68.8%	59.4% (2.1%)	+4.5	1.0×10^{-4}
Same valence, mean over groups	74.5%	63.7% (2.1%)	+5.0	1.0×10^{-4}
Fully consistent groups (of 87)	28	14.5 (3.2)	+4.2	2.0×10^{-4}

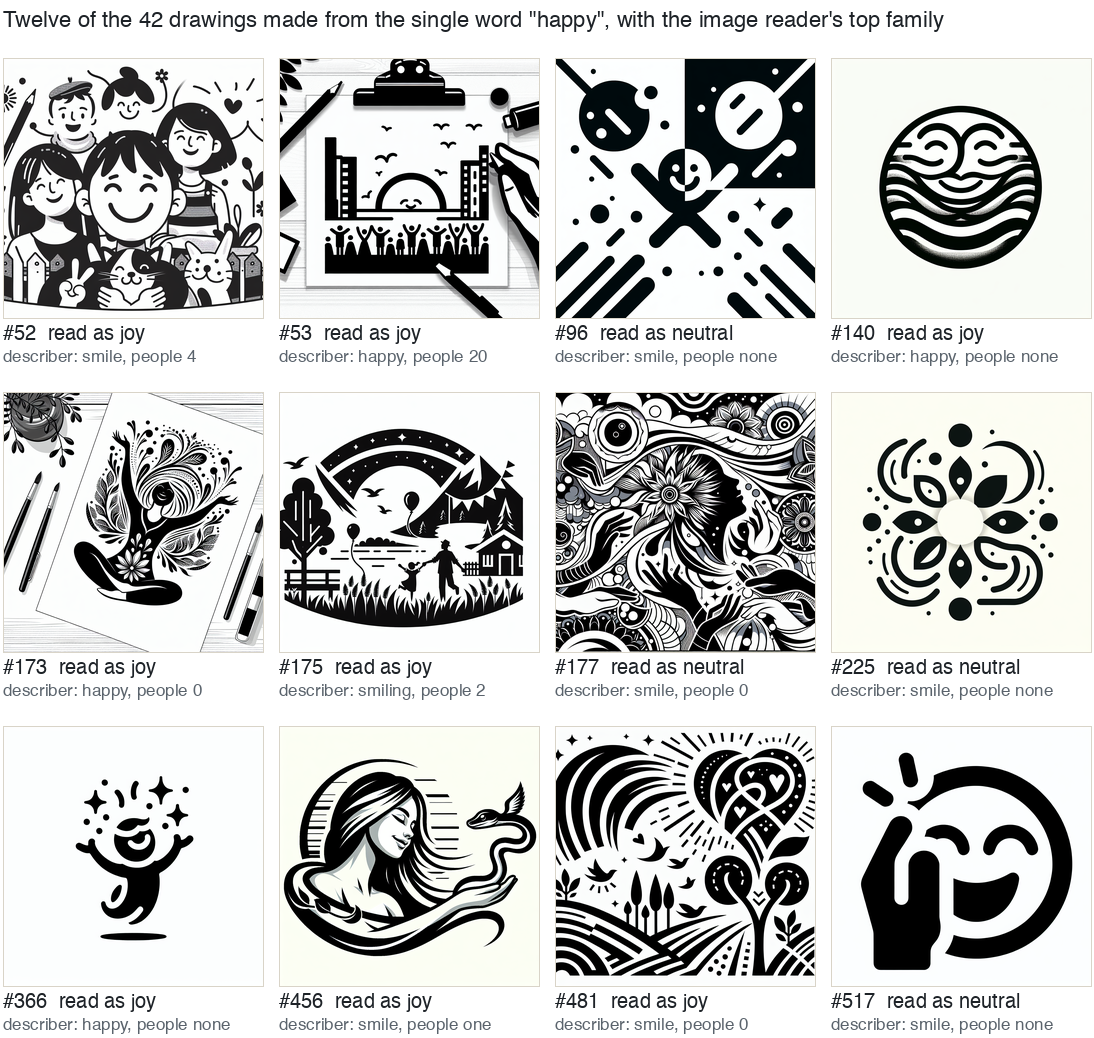


Figure 4.5: Twelve of the 42 drawings made from the prompt “happy”, each with the family the image reader returned for it.

4.7 Per-family agreement

Figure 4.6 and Table 4.9 tabulate the image reader's family against the prompt reader's for the 607 emotional records, and Table 4.11 gives the marginals of both readers over all 1,012 analysed records. Joy dominates the prompt side. It accounts for 402 of the emotional prompts, sadness for 99 and surprise for 77, while fear (16), anger (12) and disgust (1) are rare. Recall by prompt family is defined per row: of the prompts the prompt reader assigns to a family, the share whose drawing the image reader also assigns to it. Joy has recall 64.9% (CI [58.7, 70.3]) and sadness 40.4% (CI [30.9, 51.2]); surprise is at 19.5% (CI [11.4, 28.1]), anger at 50.0% on only 12 prompts and fear at 0.0% on 16. Disgust never comes back from the image reader. Fear comes back 4 times in 1,012 records.

Table 4.10 gives Cohen's kappa for family agreement. For the emotional prompts it is 0.23 (CI [0.18, 0.29]), the face-classified records give 0.30 and the scene-classified 0.13, which the Landis–Koch bands call fair, fair and slight in that order. In practice family agreement can be read on three families, joy, sadness and neutral. Anger comes fourth, and it is too rare on the prompt side, as are fear and disgust, for a claim about any one of the three.

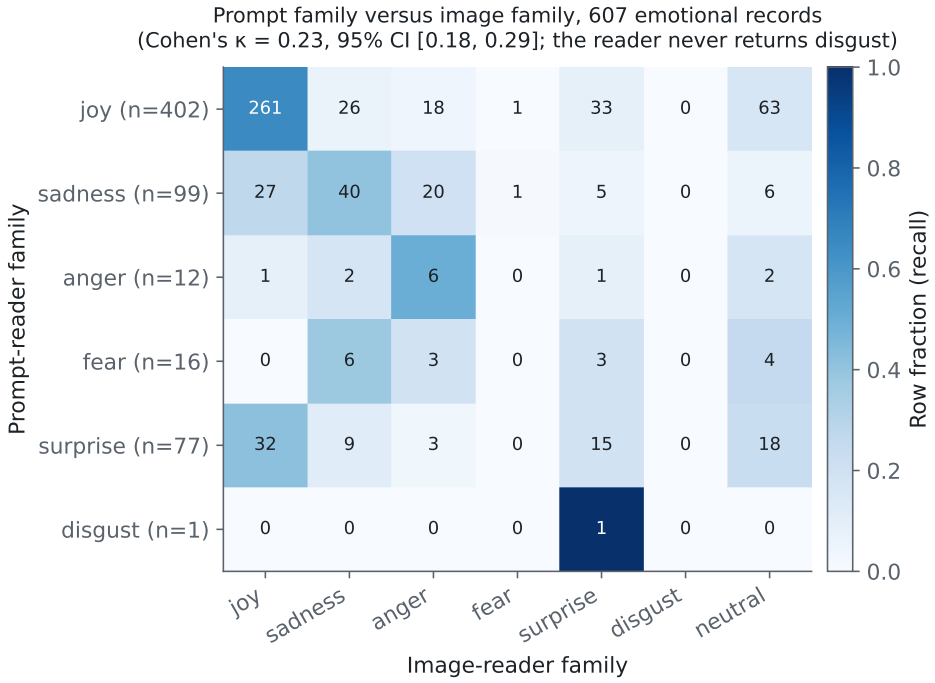


Figure 4.6: Prompt-reader family (rows) crossed with image-reader family (columns) on the 607 emotional records. Each cell holds a count, and the shading is the share of the row.

4.8 The describer

Table 4.12 tabulates, for all 1,012 records, the people count the describer gives against whether its expression answer contains an expression word, so the cells separate the records

Table 4.9: Prompt-reader family (rows) against image-reader family (columns) on the 607 emotional records. Recall = row diagonal share, with a clustered-bootstrap 95% CI. Cohen's $\kappa = 0.23$ [0.18, 0.29]. The image reader never returns disgust and returns fear 4 times in 1,012 records.

Prompt \ image	joy	sadness	anger	fear	surprise	disgust	neutral	n	Recall [CI]
joy	261	26	18	1	33	0	63	402	64.9% [59, 70]
sadness	27	40	20	1	5	0	6	99	40.4% [31, 51]
anger	1	2	6	0	1	0	2	12	50.0% [22, 78]
fear	0	6	3	0	3	0	4	16	0.0% [0, 0]
surprise	32	9	3	0	15	0	18	77	19.5% [11, 28]
disgust	0	0	0	0	1	0	0	1	0.0% [0, 0]
image total	321	83	50	2	58	0	93	607	

Table 4.10: Cohen's κ for family agreement (7 families), with clustered-bootstrap 95% CIs and the Landis & Koch (1977) band. The neutral-labelled subset is degenerate ($\kappa = 0$ by construction) and is omitted.

Subset	n	κ	95% CI	Band
emotional	607	0.23	[0.18, 0.29]	fair
face	402	0.30	[0.23, 0.37]	fair
scene	205	0.13	[0.06, 0.19]	slight
all analysed	1012	0.21	[0.16, 0.25]	fair
distinct emotional	452	0.23	[0.17, 0.29]	fair
distinct face	319	0.29	[0.21, 0.37]	fair
distinct scene	133	0.09	[0.01, 0.17]	slight

Table 4.11: Top-1 family marginals of the two readers over all 1,012 analysed records.

Family	Prompt reader	Image reader
joy	402	461
sadness	99	141
anger	12	89
fear	16	4
surprise	77	108
disgust	1	0
neutral	405	209

with a person and an expression word from those with only one of the two and those with neither. The expression question never gets a blank. For 406 records it names an expression while its own people count says nobody is there. The scene-classified records number 421, and for 414 of them the people count is zero, yet an expression word is present in 406 of those. Of the 414 records with no reported person, 332 (80.2%) give “smile”, “smiling” or “happy” as the expression. Figure 4.7 shows four such drawings with all five answers. The smile rate is given with both denominators: 66.1% over all 1,012 analysed records, 57.0% over the 591 face-classified ones.

Copy rate means the share of records on which the image reader returns the family of the describer’s own expression answer. On the strict definition the answer must be an emotion word. That is so for 418 records, 41.3% of the corpus, and for those records the family the image reader returns is the family of the describer’s own word 99.8% of the time. The broad definition also counts a facial action, “smile” or “crying” for instance, and covers 997 records. On it the copy rate is 68.6% overall, 82.2% when the describer reports a person and 48.8% when it reports none. With the “0 people” token in the description, the reading is pulled towards neutral.

Table 4.12: The describer’s answers on all 1,012 analysed records: people count crossed with whether the facial-expression answer contains an expression word. The face rule is the top-left cell. Of the 421 scene-classified records, 414 report no people and 406 of those still contain an expression word (332 answer smile, smiling or happy).

	Expression word present	Expression word absent
People ≥ 1	591	7
People $= 0$	406	8



Figure 4.7: Four scene-classified drawings for which the describer reported no person and still named a facial expression, each with its five answers as given.

4.9 Error taxonomy

The taxonomy covers the emotional records where each reader gives its top family a score of at least 0.8 and the two families differ. There are 38 of them, covering 37 distinct prompts. The intention was to sample 40, but fewer than that qualified, so every one was

examined. Figure 4.8 and Table 4.13 count the primary failure mode, which was assigned by looking at the drawing. The describer is behind 19 of the 38, 9 of those because a face or an expression it reported is not in the drawing and 10 because it misread one that is. The image reader accounts for 5, where the description was fair to the drawing and the entailment step still returned another family. 12 trace to the prompt side, 11 of them mixed or ambiguous prompts and 1 a prompt the prompt reader mislabelled, and 2 to the generator, where the drawing does not depict what the prompt says. Record 137, "i am excited", drawn as a screaming face, is one of the 2 generator cases.

The table shows two patterns. A people count of zero goes with the invented-expression mode, since 13 of the 38 records have the "0 people" tag and most of the 9 invented-expression records are among them. One mechanism accounts for the joy-to-neutral cell. "grateful" and "calm but tired" are two of the prompts in it, and both name a low-arousal positive state that none of the seven families has room for, so the describer's answer is "calm" or "smile" and the image reader returns neutral. The automatic detector for mixed prompts looked for connectives such as "but" and "although". It caught none of the 11 mixed prompts, because visitors tend to list feelings with no connective at all, as in "tired proud happy", and a detector that looks for a list would do better.

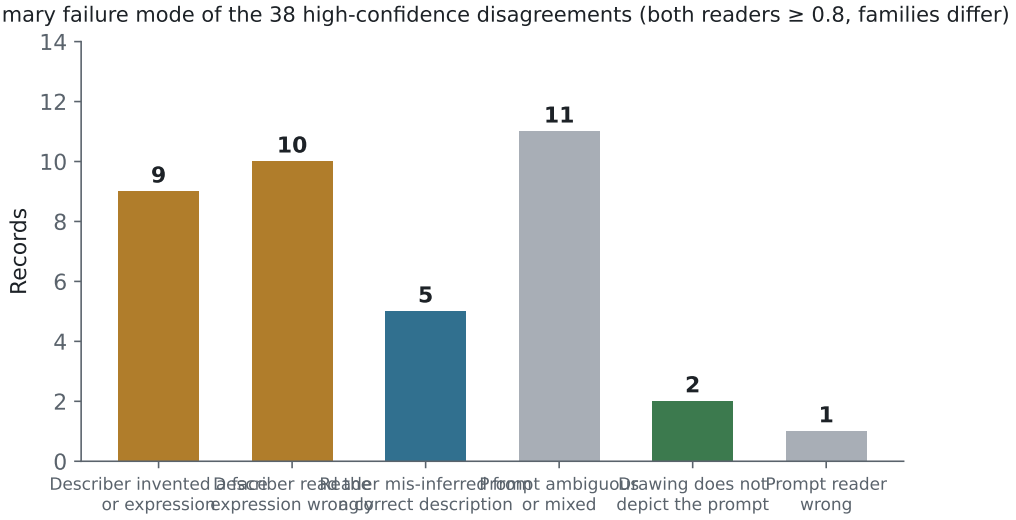


Figure 4.8: Primary failure mode of the 38 high-confidence disagreements. Each was assigned by looking at the drawing.

4.10 Description-template ablation

The template ablation is in Table 4.14 and Figure 4.9, one row per altered description template. Only the entailment stage was re-run. As a control the unmodified template was re-run through the same code path, and the number of records whose top-1 reading changed was 0, so the re-run is exact and every difference in the table comes from the template change. The unmodified row reproduces Section 4.1.

Dropping the expression sentence removes most of the agreement. For the emotional prompts, family agreement goes from 53.0% down to 20.3%, where the null is 16.6%

Table 4.13: Primary failure mode of the 38 high-confidence disagreements (both readers' top-1 score ≥ 0.8, families differ), assigned from the drawing. Describer faults (a+b) = 19; image reader (c) = 5; prompt side (d+f) = 12; generator (e) = 2.

Mode	Description	n	Exemplar ids
(a)	describer invented a face or expression where none is drawn	9	18, 19, 242
(b)	describer read the expression wrongly	10	39, 192, 202
(c)	reader mis-inferred from a correct description	5	89, 107, 231
(d)	prompt ambiguous or mixed	11	65, 66, 93
(e)	drawing does not depict the prompt	2	137, 771
(f)	prompt reader wrong	1	436

($z = +3.8$), and valence sign from 64.0% down to 15.3%, null 11.4% ($z = +5.1$). The face-classified records alone end at 18.9% on valence sign. With the expression sentence gone, the image reader's output piles up on two families, surprise on 514 of the 1,012 records and neutral on 368. Holding the expression word at "unclear" for every record collapses the ladder in the same way, with family agreement at 21.3% ($z = +3.9$). The other clauses matter less. Taking out the action clause, or the posture or scene clause, instead shifts family agreement by between one and three points, to 50.6%, 51.4% and 53.4%. Dropping the people count raises the rates and the nulls together, the scene valence-sign cell going to 63.5% against 50.9% ($z = +5.0$). Without the "0 people" token holding scene readings near neutral, the invented smile reads as joy, and joy is the top family of most emotional prompts.

Under the people-aware template the whole description becomes "A drawing showing {scene}." for any record whose people count is zero, and nothing else changes. The face rows cannot change under it. They stay at 73.9% and 61.9%, identical to the unmodified template by construction, because a face-classified record always has a person reported and so keeps its full description. The scene rows both fall to their nulls, on valence sign 4.7% where the null is 3.4% ($z = +1.6$), and on family 18.0% where the null is 17.3% ($z = +0.6$). Putting the action and posture clauses back for those records gives the scene cells 15.9% on valence sign, null 11.6% ($z = +2.7$), and 21.0% on family, null 16.8% ($z = +2.2$). Taken over every emotional prompt, the people-aware template gives 51.7% on valence sign ($z = +10.1$) and 47.1% on family ($z = +8.8$).

4.11 Single-coder reliability check

The 120 sampled drawings were coded from the contact sheets (the first is Figure 3.5), blind to the prompt, the description and both readings, marking for each one whether a person is drawn, whether a face can be read, the valence it gives and the family. Table 4.15 scores the pipeline against that coding with Wilson intervals. One person coded it once, so it is a judgement, not a reliability estimate.

The describer's people count agrees with the coder on 87.5% [80.4, 92.3] of the sample, so it can tell whether anybody is drawn. The face rule cannot. Its accuracy against the coder's judgement of whether a face is readable is 62.5% [53.6, 70.6], and it holds a recall of 77.1% [61.0, 87.9] against a precision of 42.2% [30.9, 54.4]: of the drawings it puts in

Table 4.14: Description-template ablation: the NLI stage re-run on descriptions rebuilt from the cached describer answers with one clause removed or fixed; the prompt side and the null procedure unchanged. Valence-sign and family agreement per variant on the emotional prompts and their face/scene halves.

Variant	Subset	n	Sign obs	null	z	Family obs	null	z
baseline	emotional	607	64.0%	45.6%	+11.1	53.0%	38.8%	+10.3
	face	402	73.9%	53.5%	+10.4	61.9%	45.8%	+10.3
	scene	205	42.9%	29.6%	+5.8	35.6%	26.2%	+4.4
drop_expression	emotional	607	15.3%	11.4%	+5.1	20.3%	16.6%	+3.8
	face	402	18.9%	14.7%	+3.8	25.9%	20.5%	+3.8
	scene	205	7.6%	4.7%	+2.7	9.3%	6.4%	+2.2
drop_action	emotional	607	61.9%	43.9%	+11.0	50.6%	36.7%	+10.3
	face	402	72.5%	52.2%	+10.4	60.7%	44.5%	+10.4
	scene	205	39.4%	26.8%	+5.6	30.7%	21.9%	+4.6
drop_posture	emotional	607	62.3%	44.5%	+10.7	51.4%	37.9%	+9.7
	face	402	73.1%	53.1%	+10.0	61.4%	45.6%	+10.1
	scene	205	39.4%	26.8%	+5.6	31.7%	24.0%	+3.6
drop_people	emotional	607	68.5%	50.7%	+10.8	57.8%	43.7%	+9.9
	face	402	70.8%	50.7%	+10.4	59.7%	43.6%	+10.3
	scene	205	63.5%	50.9%	+5.0	54.1%	44.5%	+3.8
drop_scene	emotional	607	64.2%	46.0%	+10.7	53.4%	39.2%	+9.8
	face	402	74.4%	53.8%	+10.4	62.4%	46.1%	+10.1
	scene	205	42.4%	29.9%	+5.6	35.6%	26.8%	+3.9
fixed_expression	emotional	607	15.1%	10.8%	+5.9	21.3%	17.5%	+3.9
	face	402	17.5%	12.8%	+4.4	25.1%	19.7%	+4.0
	scene	205	10.0%	6.6%	+2.8	13.7%	11.2%	+1.5
people_aware	emotional	607	51.7%	36.9%	+10.1	47.1%	34.5%	+8.8
	face	402	73.9%	53.5%	+10.4	61.9%	45.8%	+10.3
	scene	205	4.7%	3.4%	+1.6	18.0%	17.3%	+0.6
people_aware_no_expression	emotional	607	55.3%	39.5%	+10.5	48.1%	35.0%	+9.6
	face	402	73.9%	53.5%	+10.4	61.9%	45.8%	+10.3
	scene	205	15.9%	11.6%	+2.7	21.0%	16.8%	+2.2
control_full_template	emotional	607	64.0%	45.6%	+11.1	53.0%	38.8%	+10.3
	face	402	73.9%	53.5%	+10.4	61.9%	45.8%	+10.3
	scene	205	42.9%	29.6%	+5.8	35.6%	26.2%	+4.4

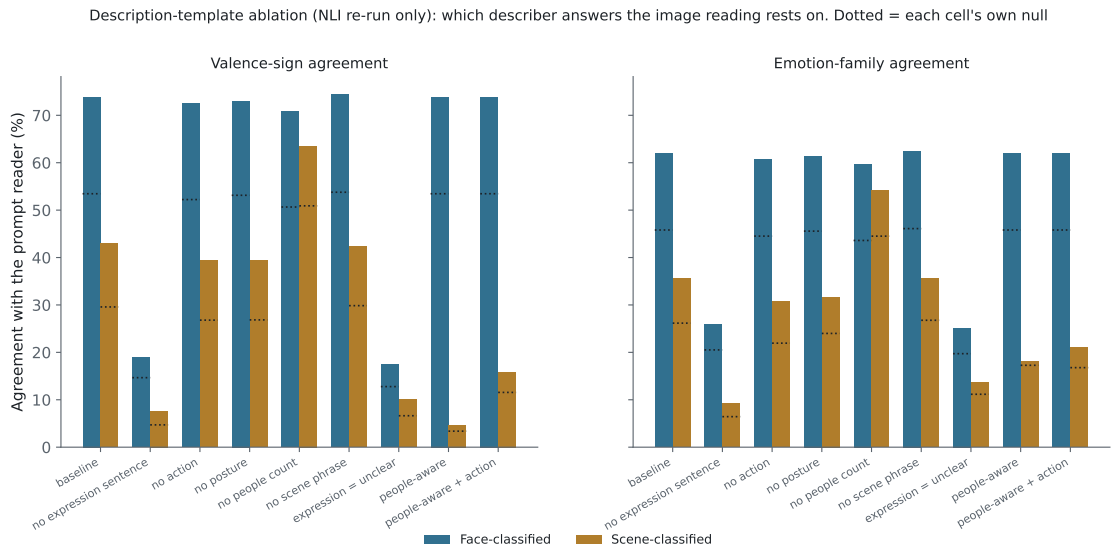


Figure 4.9: Description-template ablation with only the entailment stage re-run. Valence-sign and family agreement by image class under each template. The dotted line in each cell is that cell's random-pairing null.

the face class, fewer than half carry a face the coder could read. The "smiling" answers go the same way. Of the 77 in the sample, 18.2% [11.2, 28.2] have a readable positive face behind them and 50.6% [39.7, 61.5] have nobody drawn at all, which is the count of Section 4.8 met from the other side, by a person.

Against the coder's own reading of the drawings the image reader agrees on valence sign for 71.0% [58.7, 80.8] of the records that have a sign, and on family for 40.8% [32.5, 49.8] ($\kappa = 0.21$), the same Landis–Koch band as its agreement with the prompt reader. The last row is the one the check was built for. Splitting on the coder's classification instead of the describer's, the prompt reader's valence agrees with the coder on 100.0% [84.5, 100.0] of the drawings with a readable face and 86.4% [66.7, 95.3] of those without, a difference of +13.6 pp at $z = +1.8$. It is a different pair of readers from the +30.9 pp of Section 4.2 and cannot replace it, but it is the one split in this document that the describer had no hand in, and on it the face advantage is small and does not clear its own noise. The two rates rest on a few dozen records each, so their intervals are wide. The check passes on the people count, fails on the face rule, and is too small to settle the contrast either way.

4.12 Summary of findings

The five findings below are tied to the objectives in Section 1.3.

1. For the 607 emotional prompts, the two readers agree above chance on all four measures, with valence sign at 64.0% where chance gives 45.6% and family at 53.0% where chance gives 38.8%. Clustering, the re-run on distinct prompts and Holm's correction leave this intact (objectives 2 and 3).

2. Agreement concentrates in drawings the describer classes as faces. The

Table 4.15: Single-coder reliability check on 120 drawings coded blind to prompt and model output (Wilson 95% CIs).

Quantity	Value [95% CI]
Describer people-count accuracy	87.5% [80.4, 92.3]
Face-rule precision / recall	42.2% [30.9, 54.4] / 77.1% [61.0, 87.9]
'Smiling' answers with a readable positive face	18.2% [11.2, 28.2]
'Smiling' answers with no person drawn	50.6% [39.7, 61.5]
Image reader vs coder, valence sign	71.0% [58.7, 80.8]
Image reader vs coder, family (κ)	40.8% [32.5, 49.8] ($\kappa = 0.21$)
Prompt reader vs coder valence, readable face	100.0% [84.5, 100.0]
Prompt reader vs coder valence, no readable face	86.4% [66.7, 95.3]

face-minus-scene difference on valence sign is +30.9 pp (CI [+22.6, +40.6], clustered permutation $p = 0.0001$). The scene result shrinks on distinct prompts and drops to its null once the description stops asserting a person for drawings with none (objectives 3, 4 and 7).

3. On the neutral-labelled prompts the only measure with a live null sits at chance, 76.6% against 76.5%, and stays there once the 68 prompts with an affect word are taken out (objective 3).

4. Under like-for-like conditions, describe-then-extract and the prompt reader agree on 53.0% of families, direct CLIP and the prompt reader on 22.4%, a gap of +30.6 pp, for which the clustered sign-flip test gives $p < 0.001$ (objective 5).

5. When a prompt was drawn more than once, the image reader returned the group's modal family for 68.8% of the drawings, where chance gives 59.4% ($z = +4.5$), so the reading is steadier than chance, though well short of fixed (objective 6).

The audits bear on every item above. For 406 of the 414 scene-classified records with a people count of zero, the describer still named an expression. When the describer's expression answer is already an emotion word, the image reader returns its family 99.8% of the time, and under the ablation most of the agreement goes when the expression sentence is removed.

Chapter 5

Discussion

5.1 What the results establish

Chapter 4 reports the numbers. This chapter sets out which claims they support, where the pipeline is weakest, how much weight the face/scene contrast can bear, and where the findings stand next to the studies in Chapter 2. The results support three statements.

The first is that the prompt reader's reading of what a visitor typed and the image reader's reading of the drawing made from it agree, on all four rungs, well above the random-pairing null. Clustering on prompt groups discounts the 249 records whose prompt appeared earlier, and the intervals in Table 4.1 still clear their nulls. Keeping only the first record of each prompt lowers the rates a little and the nulls with them, and the sign z goes from +11.1 to +9.9. Holm's correction over the 36 non-degenerate comparisons keeps 26. Valence sign carries the most weight here, 64.0% where the null sits at 45.6% (z +11.1, interval [58.6, 68.8]), because it asks the simplest question and states its denominator, 530 of 607.

The second is that the agreement sits in drawings in which the describer found a face. The design set out to test faces against scenes, and the outcome is that only the face drawings agree at all for this reader. Face-classified records reach 73.9% on valence sign against 42.9% for scene-classified ones, a gap of +30.9 pp with a clustered interval of [+22.6, +40.6] and a permutation p of 0.0001. On distinct prompts the scene family cell drops to 30.8% (z +2.6) and is among the 10 comparisons Holm removes. The template ablation removes the scene-side agreement that is left. With the people-aware template the face rows are identical by construction while the scene rows fall onto their nulls, 4.7% against 3.4% on sign (z +1.6) and 18.0% against 17.3% on family (z +0.6).

The third is that the specificity check and the direct-CLIP comparison behave as expected. Where the prompt reader found no affect, the only measure with a working null sits at chance, 76.6% against 76.5% (clustered z -0.1). The control audit flagged 70 records, 68 containing a plain affect word and 2 reduced by redaction to the bare placeholder. With those removed the clustered z is -0.6. Scored like for like on the 607 records, describe-then-extract reaches 53.0% family agreement and direct CLIP 22.4%, which leaves a gap of +30.6 pp (CI [+25.5, +35.6], cluster sign-flip p <0.001). There is no ground truth, so the gap does not say which reader is the better one. The two image readers share a top-1 family on only 32.1% of these records, and describe-then-extract

agrees with the prompt reader far more often.

None of this says what a person would see, because no person has rated the drawings. Every number in Chapter 4 is one machine reader agreeing with another. The single-coder check is the only step in the project that sets a human reading beside the machine ones, and on its 120 drawings the image reader agrees with the coder's valence sign 71.0% [58.7, 80.8] of the time (Section 4.11). One coder on a sample of that size settles nothing on its own.

5.2 The describer is the weak link

All of the audits point at the describer. The description template writes a person with a facial expression into every description, whether or not the drawing has a face in it. Of the 421 scene-classified records, the describer counted nobody in 414, and for 406 of those it still named an expression. For 332 of them (80.2% of the no-person records) the word was "smile", "smiling" or "happy". Because the image reader reads only that description, the smile word goes straight into the emotion decision. Where the expression answer is a bare emotion word (418 records, 41.3% of everything analysed), the reader's top family and the family of that word coincide 99.8% of the time. On those records the image reader mostly returns the family of the describer's own emotion word, so the vision model is taking part in the emotion decision that the two-stage design was meant to keep it out of.

The ablation puts numbers on this (Table 4.14). Taking the expression sentence out of the template drops family agreement for the emotional prompts from 53.0% to 20.3%, against a null of 16.6%, and the sign rung from 64.0% to 15.3%. With the expression word pinned to "unclear" the figure is 21.3%. Removing the action, the posture or the scene clause instead moves family agreement to 50.6%, 51.4% and 53.4%, a shift of one to three points, so most of the reading rests on the expression clause.

This changes how the comparison with direct CLIP should be read. The gap of +30.6 pp may simply mean that BLIP-VQA reads a drawn face better than CLIP does, with the two stages adding little, and an ablation that strips emotion words from the expression answer before the reader gets it would tell the two apart. It has not been run. It is listed as further work in Chapter 6.

The error taxonomy points the same way (Table 4.13). Of the 38 records where both readers were confident and disagreed, the describer accounts for 19 (9 inventing a face or an expression, 10 misreading one), the image reader for 5, the prompt side for 12 (11 mixed prompts, 1 prompt-reader misreadings) and the generator for 2. So the generator is seldom the cause when the two readings differ, and one of the readers usually is. Bhattacharyya and colleagues [6] find a positive bias in vision-language models. Here it takes the form of a smile rate of 57.0% among the 591 face-classified records and 66.1% over everything analysed. Gao and colleagues [24] report that the same models mix up the feeling on a depicted face with the feeling a picture is meant to evoke in a viewer, and in this corpus that confusion appears as the 406 records where the describer named an expression but counted nobody.

5.3 Circularity of the face/scene split

The face/scene contrast was Dr Chen's hypothesis from the meeting at the end of July and the one contrast the design was built to test, so it needs its own caveats. A record is in the face class when the describer counts a person and uses a listed expression word, and in the scene class otherwise. In practice the people count does nearly all of the work, since the describer counted nobody in 414 of the 421 scene-classified records. That same description is what the image reader then reads. So the contrast compares two kinds of describer output, and its two classes are defined by the same text the reader scores.

The contrast clears its nulls, holds under clustering, and the permutation test gives $p = 0.0001$ on both sign and family. It is a narrower result than a difference between faces and scenes as drawn, because the classes come from the describer. When a person and an expression are both reported, the family the reader returns tracks the prompt; when nobody is counted, the reader is left with only the expression the template made it assert, and once the people-aware template drops that assertion the scene rows sit on their nulls. What a person would read in a scene drawing is untested. The single-coder check sorts drawings with no input from the describer, and it says the rule is worse than the contrast makes it look: its precision against the coder's judgement of a readable face is 42.2% [30.9, 54.4], so most of what it calls a face is not one, and on the coder's own split the face advantage falls to +13.6 pp at $z = +1.8$ (Section 4.11). That is a smaller sample and a different pair of readers, so it qualifies the contrast rather than overturning it. None of this was measured until the 9 September audit, though the template dated from early August.

5.4 What is not claimed

No claim is made that agreement is higher for positive prompts than for negative ones, or the reverse. My first build gave higher agreement for negative prompts than positive ones, and the second build, which had a different image reader, gave the opposite. Both were withdrawn at the audit on 1 September. Inside a single-valence subset every prompt shares one sign, so the valence-sign null has zero spread and there is no test to run, and the raw gap on the valence–arousal rung is a base-rate artefact of where the two families sit on the circumplex. The two image readers also point opposite ways on the direction, so none is reported.

The registered title names what the project set out to study, while the body measures something narrower, the agreement between two machine readers set out in Section 5.1, and the text says so wherever a number appears because the two are easy to confuse. The human rating study is described as further work in Chapter 6.

Colour is outside this study. Every drawing is monochrome and there is no colour condition. Alvarado [2] find that colour does most of the emotional work in generated images, so a monochrome corpus was expected to be a hard case, but their model uses colour features alone and says nothing about line. So nothing here can be put down to the missing colour as opposed to the readers, the family scheme or the rarity of some families.

The 68.8% within-group agreement against a null of 59.4% is a bound on the entire chain, prompt to drawing to reading, so it says nothing about the generator by itself. The

image reader's reliability was not estimated, so part of the spread inside a repeat group is measurement noise. Generation parameters are unknown as well. The files record neither seed nor rewritten prompt, so the remainder cannot be split between sampling and prompt rewriting, and a repeat group cannot tell one visitor typing twice from two visitors typing the same word.

Two restrictions remain. Fear (16 prompts), anger (12) and disgust (1) occur too seldom among the prompts for any statement at family level, disgust is never returned by the image reader, fear is returned only 4 times in 1,012 records, and surprise is mostly prompts about curiosity. Claims stop at overall agreement and at the face/scene difference. Every number is also a count of records, and the people behind the 1,012 records number at most 763, so nothing here says how many visitors felt anything.

5.5 Relation to prior work

Lomas and colleagues [49] come closest. Their secondary result, from 24 raters scoring 240 images for ten emotions, was that an emotion drawn on a person scored higher than the same emotion drawn on a robot. The same direction appears here with two machine readers and prompts that visitors typed themselves, and the ablation then shows that this version of the effect depends on the describer reporting a face, whether or not one is drawn. The two results agree in direction, but because their raters were people and these are machine readers, neither validates the other. WikiArt Emotions [59] found that images with faces get more consistent emotion judgements from annotators, and the face/scene contrast reproduces that, except that the consistency here is a machine's.

The ladder rests on Cowen and Keltner [19]. If self-reported experience falls into 27 categories joined by gradients, then a near miss between two readers is information and should be scored as such. Valence–arousal similarity sits at 83.8% against 77.3% and top-3 overlap at 50.6% against 39.4% on the same records where family agreement is 53.0%, so the two readers are near one another far more often than they land on the same family, and a one-to-one score would have discarded that closeness.

The prompt reader bounds everything downstream. The text literature says it will struggle with this register [21, 84]. Cohen's kappa for the emotional prompts is 0.23 (CI [0.18, 0.29]), in the fair band of Landis and Koch [40], which is the sort of value human annotators manage on fine-grained text emotion. GoEmotions' own annotators agreed only moderately on many of its 28 categories, and two machine readers with a generator and a drawing in between had no chance of beating that. Carrasco and colleagues [10] find that agreement between people and a generator's labels climbs steeply when the labels are reduced to positive and negative, which is the same pattern as the gap between the sign and family rungs here.

Quoting every value with its null is taken from Zhang and Wan [90] and from Lee and colleagues [41], who find that the usual image–text scores shift when an image is perturbed slightly and track human judgement weakly. A score like that should not be quoted bare. Neither should a similarity whose lowest possible value is 0.412 and which random pairings already push to about 76.6%. CLIPScore [29] is the field's default for this kind of judgement, which made direct CLIP the obvious comparison. It does poorly on these drawings because an embedding trained on captioned photographs has little to say about

a monochrome drawn face, and that is the same weakness that led the VQAScore authors [47] to a question-answering judge. On that view the describer is a VQAScore-style judge aimed at facial expression, which fits how much of the reading rests on its expression answer.

5.6 Limitations

1. Every result is agreement between two machine readers. The only human anchor is the single-coder check, and it is one coder, coding once, over 120 drawings.
2. Which prompts count as emotional is the prompt reader's call, never checked by a person; 16.8% of the control contains a plain affect word, so the control means prompts in which the reader found no affect, which is not the same as prompts without feeling.
3. Circularity affects the face/scene split, because its inputs are the describer's answers, the very text the image reader then reads, and the people count drives it.
4. For every drawing the template writes in a person and an expression. The people-aware template exists only in the ablation, so every main result uses the original one.
5. Low-arousal positive states have no family of their own among the seven, so prompts like "grateful", "calm" and "at peace" all end up as neutral.
6. The family centroids were placed by hand. Means computed per family from the NRC norms [56] would put data in place of that judgement.
7. Each z comes from 300 re-pairings and is an ordering statistic; over seeds 1–5 the headline z for valence sign moves between +11.2 and +12.2.
8. Nothing can be regenerated because the generation parameters were never recorded, so the consistency figure cannot be broken down.
9. Redaction removes only the name, so the drawing and the rest of the text stay and some re-identification risk stays with them.
10. Everything was run once, on one machine, so a second describer and a test–retest estimate for each reader are both missing.

5.7 Threats to validity

The largest threat is to construct validity, because no person has rated the drawings and agreement between two machine readers is all that is measured (Section 5.1). Two internal threats, the leak from the describer's expression answer into the image reader and the split built from the describer's own answers, were not measured until the 9 September audit (Sections 4.8 and 4.10), and the third, pseudo-replication from the 249 repeated records, is handled by the clustered inference. External validity is narrow. The corpus is one generator, one festival, mostly one language and one monochrome style, and nothing transfers to colour or to another generator without being measured again. On the

statistical side, Holm's procedure covers the 36 comparisons and the degenerate nulls are now caught from the spread of the null itself. The threat the design could not answer is the human anchor, which is the first item of further work in Chapter 6.

Chapter 6

Conclusions and Further Work

6.1 Conclusions

The project set out to measure how closely two readings of one record agree, one from what the visitor typed and one from the drawing the generator returned. There are three findings, each with the qualifications Chapter 5 puts on it. On the 607 emotional prompts the two automatic readers agree well above random pairing on each of the four rungs. That agreement is concentrated in the face-classified drawings. Once the people-aware template drops the person and the expression from the description wherever the describer counted nobody, the face-classified rows do not move and the scene-classified rows fall to their nulls, so for this reader a scene drawing gives nothing above chance.

Every objective in Section 1.3 was met. One result, the scene-classified agreement, fell to its null in the second audit, and the single-coder check then found the face rule that carries the contrast to be right less than half the time it fires. By early August the first objective, the describe-then-extract reader, had run over all 1,025 images without an unhandled error. The second objective, a random-pairing null on every rung of the ladder, did the most work, since it showed that random pairs already average 76.6% on the loosest rung. For the third, agreement measured over the corpus and on each split gives 64.0% valence-sign agreement on the emotional prompts against a null of 45.6%, 73.9% for face-classified drawings and 42.9% for scene-classified ones, with the neutral-labelled prompts at chance. The fourth objective put the face/scene gap of +30.9 pp to a clustered permutation test, which gave $p = 0.0001$, and the headline values survive the re-run on distinct prompts and Holm's correction. The scene family value on distinct prompts does not survive Holm.

In the like-for-like comparison of the fifth objective, describe-then-extract reaches 53.0% family agreement and direct CLIP 22.4% on the same 607 records. For the sixth, the mean share of a repeat group's drawings given its modal family is 68.8% over the 87 groups against 59.4% under the permutation null, so drawings of one prompt share a family more often than chance, though far from always. Under the seventh objective, the audits found that the describer named an expression for 406 of the 414 scene-classified drawings where it counted nobody. When the expression answer is a bare emotion word, the image reader returns that word's family 99.8% of the time. The ablation shows that the reading hangs on that one answer. In the error taxonomy, 19 of the 38 confident

disagreements trace to the describer and 2 to the generator. For the eighth, the code is released with its two layers of tests and the demonstrator.

Four things here are new, and none is a new metric or a new classifier. First, the corpus and the question put to it, because no earlier study was found that had short public self-reports, typed in the visitors' own words and often mixed, drawn without colour and asked whether two readings of each record agree. Second, every value is printed beside its own chance level. Third, the consistency experiment, which needs the 249 repeated records this corpus has. Fourth, the audits give an account of the instrument, which is plainer in practice than its two-stage design suggests. Although that reader was built so that no question put to the vision model mentions emotion, the audits show that for most records it amounts to a facial-expression question and a lookup from answer to family. A study of what a person sees in these drawings therefore needs human raters or a describer able to read scenes, and both are in Section 6.2.

6.2 Further work

The list runs in priority order, and each item follows from a result in Chapter 4.

1. A human-rated study, the step this pipeline was built to precede. Raters would score a stratified sample of the drawings on the same four measures, so that human and machine agreement line up rung for rung, and blind to provenance, since three studies in the review [70, 32, 54] found that knowing a machine made a picture changes the ratings. The protocol to adapt is the one Lomas and colleagues used [49], with the single-coder check as its pilot.

2. Making the people-aware template the default, on the evidence of the ablation. The pipeline and the demonstrator should switch together, so the two never drift apart, and the main table should then be re-run.

3. An expression-word ablation on the face-classified records. Take the emotion word out of the expression answer before the entailment stage and measure what is left. The gap over direct CLIP has two possible explanations, that BLIP-VQA reads a drawn face well or that the two-stage design is better whatever the describer reports, and this ablation would separate them. It has not been run.

4. A describer that reads scenes. A sketch- or drawing-specific describer was designed, but time ran out before it could be run. If raters in the human study see emotion in the scene drawings, the next describer needs to get at affect through composition, posture and setting instead of the face, and its answers would go into the people-aware template.

5. A second describer, and a test–retest estimate for both readers. Until those exist, the consistency result cannot be split into variation from the generator and noise from the measurement.

6. Per-family valence–arousal means from the NRC norms [56] in place of the hand-placed centroids, and a richer label scheme, possibly the 27 categories of Cowen and Keltner [19], so that a low-arousal positive state such as "grateful" has somewhere to go besides neutral.

7. A second coder on the same 120 drawings, with an agreement statistic between the two, so the human coding has its own reliability figure before the face/scene contrast is recomputed on it.

The most concrete step is the second coder, item 7. The 120 drawings are coded once and the sheets are ready for anybody else to code, and a second pass would give the human coding a reliability figure of its own before much weight is put on the split it produces.

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Appendices

Appendix A

Reproducibility

This appendix lists what is in the code folder, the order in which the scripts run, and the environment they ran in. Everything the dissertation quotes is produced by one of these scripts and written to a JSON file; the numbers and tables in the text are generated from those files by `export_tex.py`.

A.1 Repository layout

- `emoviz/emoviz/metrics.py`: the only definition of every measure, split rule, centroid and the random-pairing null. Every other script imports it.
- `emoviz/emoviz/data.py`: the loader. It resolves the three filename schemes, reads the signing timestamps, computes prompt groups and applies name redaction at load time.
- `emoviz/preprocess_names.py`: the three-stage redaction procedure (detect, review, apply). The reviewed decision table is kept with the dataset, not with the code.
- `emoviz/describe_extract.py`: the pipeline (BLIP-VQA, the template, BART-MNLI). It checkpoints to a JSONL file after every record and resumes from it.
- `emoviz/analysis_final.py`: the splits, the four measures, the nulls and the distinct-prompt re-run. Writes `outputs/analysis_final.json`.
- `emoviz/significance.py`: the clustered bootstrap and the clustered permutation test of the face/scene difference.
- `emoviz/inference.py`: clustered intervals for every subset, Cohen's κ, the Holm procedure, and the seed check on z.
- `emoviz/method_comparison.py`: the like-for-like comparison of the two image readers with the paired tests.
- `emoviz/data_audit_consistency.py`: the record accounting and the repeated-prompt consistency with its permutation null.
- `emoviz/audits.py`: the control audit, the describer audit and copy rate, and per-family recall.

- `emoviz/error_taxonomy.py`: the census of high-confidence disagreements and their failure modes.
- `emoviz/ablate_nli.py`: the description-template ablation (only the entailment stage re-runs).
- `emoviz/human_check.py`: draws the single-coder sample, lays out the contact sheets, and scores the coded file.
- `emoviz/make_figures.py` and `emoviz/export_tex.py`: every figure, and every number and table in this document, from the JSON files. The four extra figures in this document are made by `dissertation/build_notes/make_extra_figures.py`.
- `emoviz/app.py` and `emoviz/web/`: the demonstrator.
- `emoviz/test.sh` and `emoviz/test_numeric.py`: the smoke suite and the pinned numeric checks.
- `emoviz/outputs/`: cached model outputs, the result JSON files, the figures, and the frozen direct-CLIP file.

A.2 Run order

1. `bash emoviz/setup.sh` creates the environment from the pinned `requirements.txt`.
2. `describe_extract.py` runs the pipeline over all images. Its output is cached, so this step is only repeated to regenerate from scratch.
3. `analysis_final.py` runs the main analysis from the cached outputs in a few seconds.
4. `significance.py`, `inference.py` and `method_comparison.py`.
5. `data_audit_consistency.py`, `audits.py`, `error_taxonomy.py` and `ablate_nli.py` (the ablation takes about half an hour because it re-runs the entailment model).
6. `human_check.py sample`, then code `outputs/human_check/labels.csv` from the sheets, then `human_check.py score`.
7. `make_figures.py` and `export_tex.py`, then build this document with `pdflatex`, `bibtex`, and `pdflatex` twice more.
8. `bash emoviz/test.sh` runs everything above that is fast, plus `test_numeric.py`.

All scripts are run with `emoviz/.venv/bin/python` from the `emoviz/` directory. The dataset folder sits next to `emoviz/` and is never copied or uploaded.

A.3 Environment

Python 3.12 with PyTorch 2.13 and Transformers 5.14, on an Apple-silicon laptop with 16 GB of memory; the models run on the Metal backend in half precision. The models are `SamLowe/roberta-base-go_emotions` (about 125 million parameters), `Salesforce/blip-vqa-base` (about 385 million), `facebook/bart-large-mnli` (about 407

million) and OpenCLIP ViT-B/32 with LAION-2B weights for the comparison reader. The first run downloads about 3.5 GB of weights. The pipeline processed all 1,025 images without an unhandled error, which is a statement about robustness and not about correctness; the per-record time was between about one and twelve seconds, slowing when memory was short. Every random process is seeded (the analysis seed is 7).



Figure A.1: The automated self-test after a full run (`bash emoviz/test.sh`): the environment and data checks, every analysis script run to a clean exit, the pinned numeric regression, the live pipeline over three fresh images, the redaction audit, and the figures rebuilt from the result files.

Appendix B

Instruments

This appendix gives the fixed strings and tables the pipeline runs on, so that any number in the dissertation can be traced to the exact question, template or mapping that produced it.

B.1 The five describer questions

The describer is asked these questions in this order, and each answer is stored per record:

1. *What is the facial expression?* (`expression`)
2. *What is the person doing?* (`action`)
3. *What is the body posture?* (`posture`)
4. *How many people are there?* (`people`)
5. *What is in the picture?* (`scene`)

None of the questions contains the word "emotion".

B.2 The description template

The five answers are put into one sentence:

```
A drawing of {people} showing {scene}.  The person is {action},
{posture}.  The facial expression is {expression}.
```

The template is applied to every record, including those where `people` is "0" or "none". The ablation variants in Section 4.10 drop one clause at a time, fix `expression` to "unclear", or (the people-aware variants) reduce the sentence to "A drawing showing {scene}." wherever no person was reported.

B.3 The image reader's hypothesis

The image reader scores each family by entailment against the hypothesis

```
This scene conveys {family}.
```

for the seven families in turn, and the three highest entailment scores are kept.

B.4 GoEmotions categories to families

joy admiration, amusement, approval, caring, desire, excitement, gratitude, joy, love, optimism, pride, relief

sadness disappointment, embarrassment, grief, remorse, sadness

anger anger, annoyance, disapproval

fear fear, nervousness

surprise surprise, realization, confusion, curiosity

disgust disgust

neutral neutral

A family's score is the sum of its members' probabilities, and the seven sums are normalised to a distribution. The grouping is the one proposed by the GoEmotions authors.

B.5 Family centroids on the circumplex

Family	Valence	Arousal
joy	+0.80	0.60
sadness	−0.75	0.30
anger	−0.70	0.80
fear	−0.65	0.75
surprise	+0.30	0.80
disgust	−0.70	0.60
neutral	0.00	0.30

The centroids were set by hand from the usual placement of the basic emotions on the circumplex. The smallest value of Equation 3.1 over these seven points is 0.412 (joy against sadness).

B.6 Readable expression words

A record is face-classified when the describer reports at least one person and its expression answer contains, as a whole token, one of:

> smile, smiling, happy, sad, angry, serious, surprised, surprise, calm, excited, crying, laughing, frowning, scared, shocked, worried, concentration, concentrating, grin, grinning, neutral, confused, determined, pensive, sleepy, sleeping.

The people count is parsed from digits or number words (none, one, two, …, several, many, crowd, group, lots). A non-empty answer that is neither is taken to mean at least one person is present.

B.7 Name redaction procedure

1. **Detect.** Candidates come from three detectors: a given-name gazetteer (7,944 names), mid-sentence capitalised tokens that are absent from a 236,000-word English dictionary, and naming phrases such as "called X" and "my friend X". Recall matters more than precision here because most prompts are typed in lower case.
2. **Review.** Every candidate has a recorded decision (redact or keep) with a reason, and each redaction is tied to the record it was reviewed in. The audit fails if any candidate has no decision or if a redaction entry no longer matches anything.
3. **Apply.** At load time, per record, the token is replaced by `[name]`; consecutive placeholders collapse to one. The files on disk are never changed. Words about illness, grief and places are kept.

B.8 Single-coder check: coding instructions

The coder works from numbered contact sheets that show only the drawings and fills four columns per drawing:

person y / n: is at least one person or face drawn?

face y / n: is there a face whose expression can be read? (n if there is no person)

valence + / − / 0: the feeling the drawing gives, positive, negative or neither

family joy / sadness / anger / fear / surprise / disgust / neutral

The sample is fixed by seed: 120 drawings, made up of 40 face-classified records, 40 scene-classified records with no people reported, and every record whose people answer was vague or a bare word, topped up equally from the first two strata. The prompt, the description and the model readings are not shown during coding.

Appendix C

Supervision record

Supervision ran roughly fortnightly from June, with a written brief prepared before each meeting. The table lists the meetings and what each one changed in the project. The direction given at the end of July shaped the whole design and is described in Chapter 3.

Date (2026)	What was discussed and what followed
18 June	First meeting. Get to know the dataset, start reading around text-to-image generation, get the School template.
9 July	The two strands of the project agreed: emotion from the text, and analysis of the drawings. A six-part structure for the write-up (abstract, introduction, literature review, methodology, results, discussion), and the instruction to argue from cited evidence and not from opinion. Dataset findings presented alongside: the three filename schemes, the timestamps recovered from the image files, the thirteen pre-festival records, the repeated prompts; questions raised about the original prompt sent to the generator and about names in the data.
31 July	The methodological direction, and the timeline for the remaining work: measure how similar the two readings are instead of scoring matches; describe the drawing first and extract the emotion from the description; split faces from scenes and emotional from non-emotional prompts; redact names and keep illness content; build in an evaluation.
20 August	The demonstrator shown running live. Timing of a human-rated study discussed; agreed to write up the computational results and treat the rated study as the next step.
4 September	Project presentation, with a live demonstration.

Appendix D

Glossary

Record One prompt paired with the drawing made from it. Records are not linked to people. 1,025 were processed and 1,012 analysed.

Prompt The words a visitor typed. Prompts are quoted exactly as typed, lower case included.

Drawing The black-and-white line drawing the installation returned for a prompt.

Prompt reader The text model that reads the prompt: a RoBERTa model fine-tuned on GoEmotions (28 categories), summed into seven families.

Describer The vision model that answers five factual questions about a drawing (BLIP-VQA). It is never asked for an emotion.

Image reader The text model that infers a family from the describer's one-sentence description (BART-MNLI, zero-shot).

Describe-then-extract The image side of the pipeline: describer, template, image reader.

Direct CLIP The comparison reader, an OpenCLIP model asked for a family directly. Called the baseline; nothing else is.

Category One of the 28 GoEmotions labels.

Family One of seven groups the categories are summed into: joy, sadness, anger, fear, surprise, disgust, neutral.

Agreement The quantity this dissertation measures: how similar the prompt reader's reading of the words is to the image reader's reading of the drawing. It describes two machine readers and says nothing about what a person would see.

Valence sign + for joy; − for sadness, anger, fear and disgust; 0 for surprise and neutral. Valence-sign agreement is computed over records whose prompt family has a sign.

Family agreement The share of records on which the two readers' top families coincide.

Top-3 soft overlap The overlap of the two readers' top-3 distributions once each is normalised to sum to one.

Valence–arousal similarity One minus the normalised distance between the two readings' score-weighted centroids on the circumplex. Its floor over the family centroids is 0.412, and randomly paired records already average about 76.6%.

Random-pairing null The mean and spread a measure takes when the image readings are re-paired with the prompts at random (300 re-pairings, seed 7). Every reported value is read against it.

z (observed − null mean) / null standard deviation. Used to order effects, not as a precise probability: across seeds it moves by about a unit (+11.2 to +12.2 for the headline).

Degenerate null A null with zero spread. It arises when every prompt in a subset shares one label, so re-pairing cannot change the score. No z is reported for it.

Prompt group All records whose prompt text is identical after case-folding, computed before redaction. There are 763 groups; 249 records repeat an earlier prompt. Every interval and test clusters on the group.

Repeat group A prompt group with two or more records (87 of them, 336 records).

Clustered bootstrap Resampling prompt groups, with replacement, to get 95% confidence intervals that respect repeated prompts (10,000 draws).

Face-classified / scene-classified A record is face-classified when the describer reports at least one person *and* its expression answer contains a readable expression word. Every other record is scene-classified. The split is the describer's, not a person's.

Emotional / neutral-labelled prompt A prompt is emotional when the prompt reader's top family is not neutral (607 records); otherwise it is neutral-labelled (405). The label was never checked by a person.

Specificity check The expectation that neutral-labelled prompts show no agreement above their null. Only valence–arousal similarity can move there.

Copy rate Where the describer's expression answer is itself an emotion word, the share of records in which the image reader's top family is that word's family (99.8% on the strict definition).

Consistency For repeat groups, the mean share of drawings in a group that the image reader gives the group's most common family, read against a permutation null.

People-aware template The ablation variant of the description template that leaves out the person and expression clauses wherever the describer reported no people.

Single-coder check The 120-drawing sample the author codes blind, from contact sheets, to score the describer and the image reader against a person.

Demonstrator The local web application that runs the real pipeline on any record or an uploaded image and shows every measure against its chance level.